Analysis

Assessment of computational methods for diabetic retinopathy lesion segmentation in fundus images

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Pixel-level segmentation of microaneurysms, hemorrhages, hard exudates and soft exudates makes automated grading of diabetic retinopathy interpretable, and dozens of methods report results on the same few public datasets. Whether those numbers can be compared is unclear: splits, input resolution and metric implementations differ, and many baselines are copied from earlier papers. Here we assessed 39 model variants on 8 test sets from seven datasets (5,121 images) under one protocol. Of 21 fundus-specific methods, 4 came with training code and 2 with weights; a widely reused 11-threshold implementation of the area under the precision–recall curve scored a released checkpoint at 0.676 where the exact value was 0.639, and the released training code, run with the authors' settings on three seeds, gave models 5–7 points below the released weights. After unified retraining of 19 models, two adapted foundation models and a general-purpose high-resolution network led both internal tests by margins within the seed-to-seed variation, a third adapted foundation model, a Swin transformer and the self-configuring nnU-Net followed within 0.04, the seven fundus-specific networks were below them, and 11 models fell within the critical difference over eight test sets. Outside the training source, performance followed the annotation style and field of view of the test images: models trained on DDR held their score on pixel-labelled standard-field sources and lost a third to a half of it on coarsely annotated, region-labelled and ultra-widefield images, models trained on the 44 IDRiD images lost on every other source, and on the coarse, region-labelled and ultra-widefield images models from the lower half of the internal tables took the lead. Training source, annotation granularity and input resolution moved scores as much as or more than architecture did; under a shared decoder, ophthalmic pretraining of the encoder added nothing, and natural-image ViT-L encoders gave the highest single-model scores of the study; averaging the maps of several models gained 0.02–0.06 on fine-grained test sets. We release code, splits and probability maps as a benchmark.

Diabetic retinopathy (DR) is graded from lesions that are visible in colour fundus photographs: microaneurysms (MA), hemorrhages (HE), hard exudates (EX) and soft exudates or cotton wool spots (SE). A model that outlines these lesions pixel by pixel shows a grader why an image received its grade, supports screening programmes in which the decision must be auditable, and provides quantitative lesion burden for follow-up[1,2]. Public datasets with pixel-level labels, first IDRiD[3,4] and DDR[5] and later FGADR[6], Retinal-Lesions[7] and TJDR[8], have turned the task into a standard benchmark, and methods built for it now span multi-scale convolutional networks[9–12], relation and transformer designs[13,14], semi-supervised schemes[15,16] and adaptations of segmentation foundation models[17–20].

Several properties of the task make the reported progress hard to read. (1) The datasets are small and heterogeneous. IDRiD has 81 annotated images, its official test set 27, and soft exudates appear in about half of them; other sources were annotated with ellipses and polygons by many graders, or as regions with confidence levels by four experts[21], so that "ground truth" means different things from one source to the next. (2) The lesions are small. A microaneurysm spans about half a percent of the field-of-view diameter, and shrinking an image to the 224-pixel input of a common backbone removes 34.3% of the microaneurysms and 35.3% of the hard exudates from the DDR training labels before any model is trained. (3) Evaluation is not standardized. Papers differ in the split (official or their own), the resolution at which predictions are scored, whether pixels outside the field of view count, and how the area under the precision–recall curve (AUPR) is computed; a widely reused implementation samples the curve at 11 thresholds. (4) Baseline numbers travel between papers without being rerun, sometimes across several generations and across different splits. (5) Claims about foundation models mostly come from the papers that propose an adaptation, with zero-shot use, prompting from ground truth and fine-tuning reported side by side.

Earlier syntheses do not resolve these issues. Surveys collect the numbers that papers report[1,2]. The closest study, by Playout and Cheriet[22], trained U-Net-type baselines across five datasets and showed that annotation style governs cross-dataset transfer; it did not rerun fundus-specific methods, did not audit published numbers and did not include foundation models. Outside ophthalmology, standardized reassessments have repeatedly found that part of the reported progress disappears once baselines are configured with care[23,24] and that rankings on small test sets are fragile[25]. A recent assessment of T cell receptor–epitope predictors[26] supplies the design we follow: evaluate the original models, retrain everything that can be retrained under one protocol, and judge both on independent data.

We therefore assessed methods for DR lesion segmentation along five lines. (1) Systematic collection: a fixed literature search and a ledger of 86 records, 8 test sets with frozen splits, and 39 registered model variants with a reproducibility level for each. (2) Audit of published numbers: which split, which metric implementation and which provenance stand behind each reported value. (3) Original models: released checkpoints scored with the authors' metric and with ours, and the authors' pipelines rerun from scratch. (4) Unified retraining: one protocol for data, training amount, augmentation, checkpoint selection and evaluation, up to three seeds, and six test sets from five sources that were never used for training.

Table 1 Datasets, frozen splits and roles. All eight sources carry pixel- or region-level labels for microaneurysms (MA), hemorrhages (HE), hard exudates (EX) and soft exudates (SE). Granularity decides which test sets may be averaged: averages are taken only within one tier. Split is train / validation / test in images; official test sets are never changed. "Maps" marks sources whose probability maps we can release.

Source	Ref.	Images	Resolution	FOV	Annotation	Granularity	Split	Maps
Training sources (internal test = official test set)								
IDRiD	[3]	81	4288×2848	50°	pixel contours, expert review	Fine	44 / 10 / 27	✓
DDR	[5]	757	variable	45°	pixel masks	Fine	383 / 149 / 225	✓
Independent test sources (never used to train a main-table model)								
TJDR, standard field	[8]	257	2048×2048	35–50°	polygons, 2 graders + adjudicator	Fine	172 / 30 / 55	✓
TJDR, ultra-widefield	[8]	304	3912×3912	133°	polygons, 2 graders + adjudicator	Fine	209 / 37 / 58	✓
MAPLES-DR	[42]	198	1440×960–2304×1536	45°	pixel masks, 7 retinologists; MA/HE/EX AI-assisted	Fine	– / 0 / 60	✓
FGADR	[6]	1,842	1280×1280	n.s.	pixel masks, 2 residents + senior	Mixed	1,096 / 193 / 553	✗
Retinal-Lesions	[7]	1,593	896×896	n.s.	ellipses and polygons, ≥3 of 45 graders	Coarse	951 / 164 / 478	✗
DiaRetDB1	[21]	89	1500×1152	50°	regions with confidence, 4 experts	Region	– / 28 / 61	✓

FOV, field of view; n.s., not stated in the source paper. IDRiD: 10 of the 54 official training images are held out for validation in unified retraining; author-recipe retraining uses all 54. DiaRetDB1: ground truth is the mean confidence of four experts thresholded at 0.75; the 28 official training images are used only to choose thresholds. Retinal-Lesions: split by patient; grey-level 127 regions are ignored. MAPLES-DR: labels drawn at 1500 × 1500 on the field-of-view crop of MESSIDOR images and placed back at the original resolution; the 138 training images are not used, the official 60 test images are; uncertain red (MA or HE) and bright (EX, SE or drusen) lesions are ignored for the two classes they could belong to. The FGADR licence forbids redistribution of derived data; maps for Retinal-Lesions are not released either.

(5) Factors: annotation granularity, training source, training amount, input resolution, lesion size, loss and the way a foundation model is used, each varied alone. The study answers three questions: how many published numbers can be reproduced and where the gaps come from (Q1); which family of methods is better once conditions are equal, and whether the answer survives on independent data (Q2); and which properties of the data and the protocol decide performance (Q3).

Results

Data collection and study design

We designed a workflow that joins data collection, model collection and four kinds of evaluation (Fig. 1a). Eight test sets were prepared from seven datasets, 5,121 images in total (Fig. 1b and Table 1). IDRiD and DDR are the training sources of the main tables, because both have official splits and almost every published number refers to one of them. TJDR, split into its standard-field and 133° ultra-widefield parts, MAPLES-DR, FGADR, Retinal-Lesions and DiaRetDB1 are independent sources: no model in the main tables was trained on them, and every checkpoint is applied to their test sets as it is. The sources differ in how lesions were drawn. Following the analysis of lesion count and lesion area by Playout and Cheriet[22] we assigned each source to one of four granularity tiers, fine, mixed, coarse and region consensus, and we average results only within a tier.

A fixed search of three bibliographic databases identified 1,389 records; 764 remained after a Boolean gate and de-duplication, 112 entered the full-text queue, and screening of the priority list added 24 methods and 2 datasets to a snowballed corpus, for a ledger of 86 records (Fig. 1d). From it we registered 39 model variants in five families and recorded for each a reproducibility level from R0 (no code) to R4 (training code, weights, splits and environment) (Fig. 1c and Table 2). 22 variants form tier 1 and enter the main comparisons: general segmentation networks (U-Net[27], DeepLabv3[28,29], HRNet[30], Swin-Unet[31], FCT[32], H2Former[14]), the self-configuring nnU-Net[23], seven fundus-specific networks (M2MRF[10], HACDR-Net[11], HRDecoder[33], PMCNet[12], WSRFNet[34], CLC-Net[35] and TC-Net[36]; the last five added to the registry on 14 October 2026 so that the fundus-specific family is more than two models), and eight foundation models (SAM[17], SAM 2[18], SAM 3[37], MedSAM[19], SAMed[38], SAM-Adapter[39], SAM2-UNet[40] and RETFound[41]). 6 variants in tier 2 are added as time allows and 11 in tier 3 are audited from their papers only.

Each model can appear under four evaluations that are never mixed in one column. The paper audit records the numbers as published and where they came from. Author weights are released checkpoints scored on official and independent test sets. Author recipes are the authors' pipelines rerun from scratch. Unified retraining applies one protocol, called the unified protocol below, to the tier-1 models with complete training code. Two test scenarios are reported throughout, as in the template study[26]: the internal test, which is the official test set of the training source, and the independent test. The primary metric is the exact AUPR of each lesion class over all field-of-view pixels of a test set, averaged over the four classes (mAUPR); Dice, intersection over union and lesion-level recall by lesion size are secondary. Finally, the factors of Fig. 1e are varied one at a time on one to five representative models.

The numbers in the literature

We first asked what stands behind the numbers that papers report. Of the 21 fundus-specific methods in the registry (18 dedicated architectures and 3 adaptations of foundation models to this task), 6 have any author code, 4 have training code and 2 (M2MRF and HRDecoder) come with weights, splits and an environment description (Fig. 2a). Authors of methods without code or weights were contacted with a fixed template; methods that could not be run stay in the denominator.

For every method paper in the ledger (29 method papers and one prior cross-dataset benchmark[22]) we extracted the

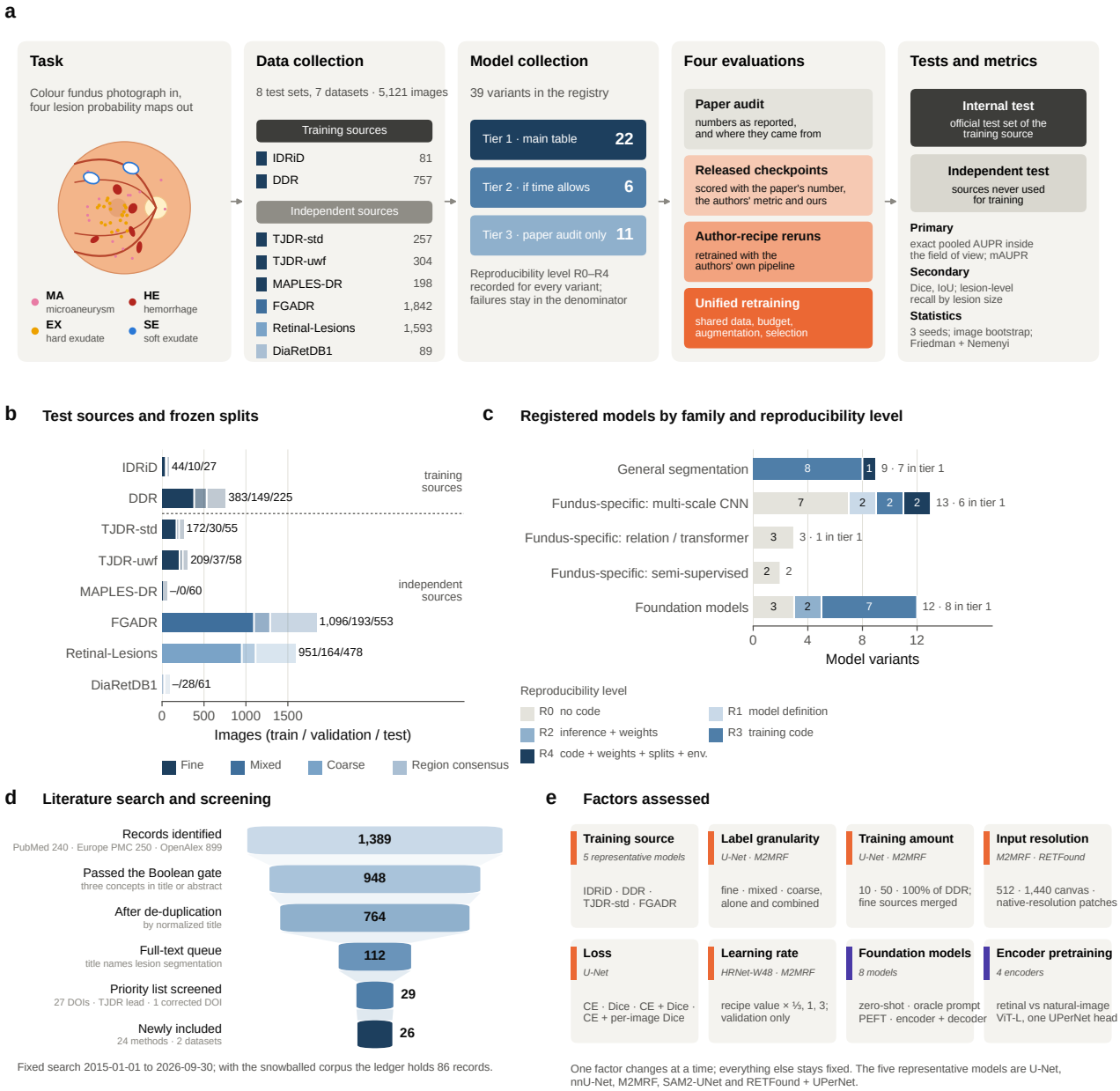


Fig. 1 Data collection and study design. **a**, Workflow: the task, the eight test sources, the registered model variants, the four kinds of evaluation, and the two test scenarios with their metrics. **b**, Images per source and frozen split (train / validation / test). Colour gives the annotation granularity tier; shade separates the three parts of a split. **c**, Registered model variants by family and reproducibility level; the number of tier-1 variants is written after each bar. **d**, Literature search and screening. **e**, Factors varied one at a time, with the models used for each.

Table 2 Models entering the retraining tracks. 39 variants are registered: 22 in tier 1 (main tables), 6 in tier 2 (if time allows) and 11 in tier 3 (paper audit only; Supplementary Note 1). Reproducibility level: R0, no code; R1, model definition only; R2, inference code and weights; R3, training code; R4, training code, weights, splits and environment. Tracks: A, author weights; B, unified retraining of the official code; C, re-implementation from the paper, reported in a separate column.

			Track				
Model	**Tier**	**Reproducibility**	**A**	**B**	**C**	**Code status**	**Use in this study**
General segmentation and self-configuring baselines							
U-Net	1	●●●○ R3	✗	✓	✗	reference implementation	unified retraining
DeepLabv3	1	●●●○ R3	✗	✓	✗	torchvision, ResNet-101	unified retraining
HRNet-W48	1	●●●○ R3	✗	✓	✗	official, W48	unified retraining
nnU-Net	1	●●●● R4	✗	✓	✗	official, 2D; default and ResEnc-M	own entry point
Swin-Unet	1	●●●○ R3	✗	✓	✗	official	unified retraining, 224 patches
FCT	1	●●●○ R3	✗	✓	✗	official PyTorch port	unified retraining, 384 patches
H2Former	1	●●●○ R3	✗	✓	✗	official	deferred: exceeds 40 GB at 960
UNet++	2	●●●○ R3	✗	✓	✗	official	tier 2
UPerNet	2	●●●○ R3	✗	✓	✗	official	tier 2
Fundus-specific, multi-scale CNN							
M2MRF	1	●●●● R4	✓	✓	✗	official; weights released	released weights; author-recipe rerun; unified retraining
HACDR-Net	1	●●●○ R3	✗	✓	✗	official; no weights	unified retraining
HRDecoder	1	●●●● R4	✓	✓	✗	official; weights released	released weights; author-recipe rerun; unified retraining
WSRFNet	1	●○○○ R1	✓	✓	✗	model and test code only; training loop added	unified retraining (weights requested)
PMCNet	1	●●●○ R3	✗	✓	✗	official Keras code rewritten in PyTorch	unified retraining
CLC-Net	1	○○○○ R0	✗	✗	✓	no code; re-implemented from the paper	unified retraining (re-implementation)
L-Seg	2	●○○○ R1	✗	✗	✓	Caffe prototxt and loss layer	C
SAA	2	○○○○ R0	✗	✗	✓	repository empty	C
Fundus-specific, relation or transformer							
TC-Net	1	○○○○ R0	✗	✗	✓	no code; re-implemented from the paper	unified retraining (re-implementation)
RTNet	2	○○○○ R0	✗	✗	✓	no code; image + vessel input	C
Foundation models							
SAM	1	●●○○ R2	✗	✓	✗	official inference	zero-shot, oracle prompt
SAM 2	1	●●●○ R3	✗	✓	✗	official	zero-shot, oracle prompt
SAM 3	1	●●●○ R3	✗	✓	✗	official; gated weights	text prompt
MedSAM	1	●●●○ R3	✗	✓	✗	official	oracle box, full fine-tuning
SAMed	1	●●●○ R3	✗	✓	✗	official	PEFT (LoRA)
SAM-Adapter	1	●●●○ R3	✗	✓	✗	official	PEFT
SAM2-UNet	1	●●●○ R3	✗	✓	✗	official	PEFT
RETFound + UPerNet	1	●●●○ R3	✗	✓	✗	official encoder; UPerNet head added	encoder + decoder
PerSAM	2	●●○○ R2	✗	✓	✗	official inference	one-shot, tier 2

general segmentation self-configuring fundus-specific foundation model. Unified retraining, the shared protocol (Methods). PEFT, parameter-efficient fine-tuning. Status as of the code audit of 2–4 October 2026; author replies are due by 31 October 2026.

experimental setup and the comparison tables, 1,538 baseline AUPR values on IDRiD and DDR in all (Supplementary Tables 1 and 2; Supplementary Data 2). The official split is the rule: 27 of the 30 papers that use IDRiD and 20 of the 24 that use DDR state it, and the others give no split or merge the official test images into training. What the papers leave out is how the metric was computed. Of 27 papers whose main metric is AUPR, 18 give no implementation, 6 integrate over a fixed grid of 11 or 33 thresholds, which is the IDRiD challenge protocol carried on by the evaluation code released with L-Seg and M2MRF, 2 use the exact average precision and 1 averages over images; none states that the evaluation is restricted to the field of view, and 5 say whether pixels are pooled over the test set or scores are averaged over images (Fig. 2b). Baseline numbers travel between papers. Of the baseline values, 28% were produced by the paper that reports them, 22% are identical to the value in the method's own paper, 26% are identical to a value that an earlier paper had already copied or produced, and 23% cite a source outside the ledger or none. Only 5 papers state that every baseline was rerun, and 10 say nothing about where their comparison numbers come from. As a consequence one method carries a range of values: L-Seg is reported on IDRiD by 14 papers with mAUPR from 0.40 to 0.66, DeepLabv3+ by 11 papers from 0.48 to 0.67 and U-Net by 6 papers from 0.35 to 0.65. Copied values agree to the last digit; the spread comes from the papers that reran a method under their own settings (Fig. 2c). The audit also surfaced copied tables whose lesion columns are transposed with respect to the source, a baseline table reproduced verbatim from an earlier paper of the same group without attribution, and a Dice value carried into an AUPR column (Supplementary Data 2).

The metric implementation alone moves the numbers. The evaluation code released with M2MRF, which later work reuses, computes AUPR from 11 thresholds; at threshold 0 every pixel with a non-zero probability is counted as positive, so the recall is 1 and the precision equals the prevalence, which lifts the area under the curve. On the released IDRiD checkpoint this function returns 0.676; after the probabilities are stored as 16-bit maps the same function returns 0.647, and the exact pooled AUPR inside the field of view is 0.639. On DDR the three values are 0.492, 0.426 and 0.401 (Fig. 2d). Applied to the probability maps of the retrained models, the 11-threshold implementation changed mAUPR on IDRiD by between −0.010 and +0.023 for the converged models (and by +0.060 for FCT), but on DDR, where the low-probability tail of the maps carries more weight, it added between 0.004 (TC-Net) and 0.050 (U-Net) for the converged models,

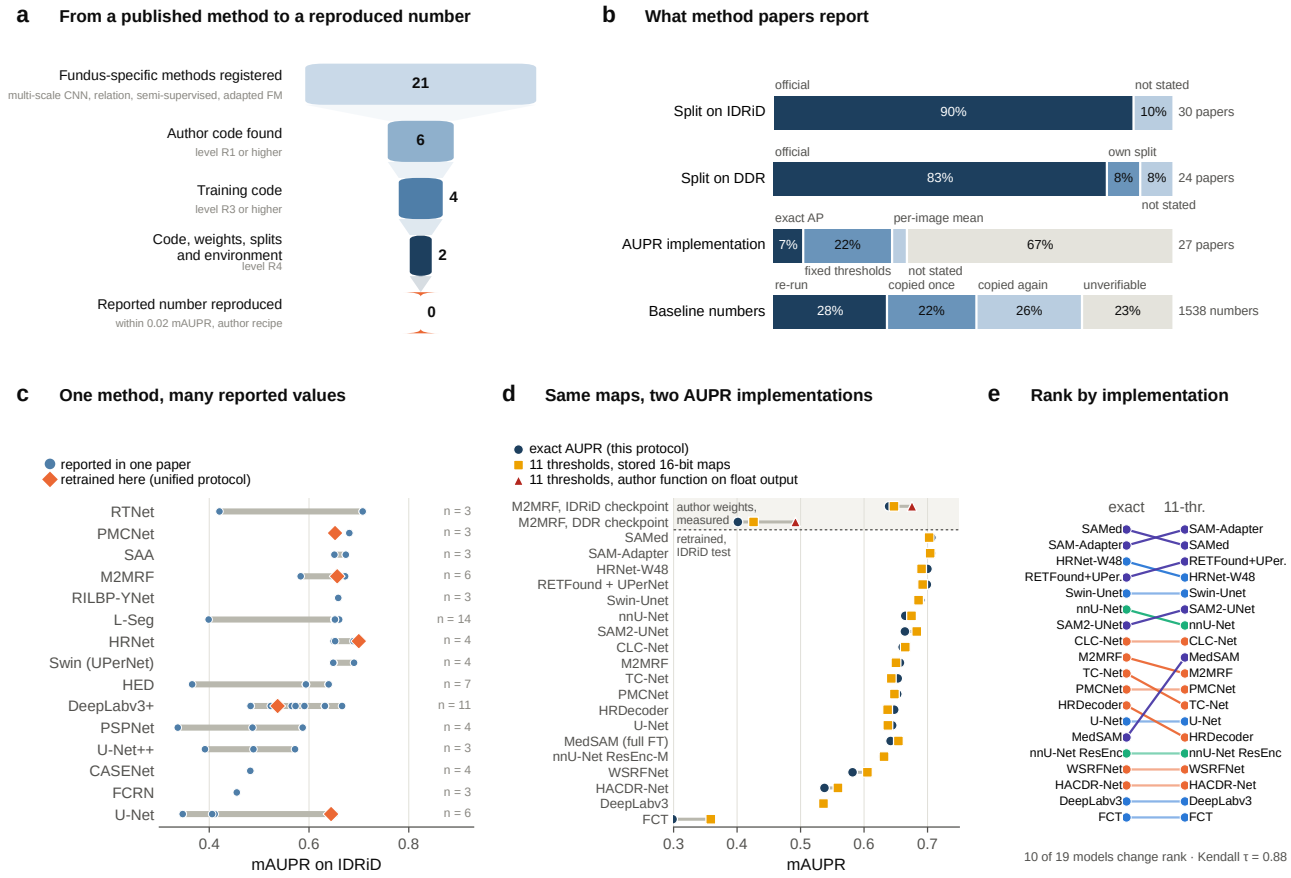


Fig. 2 The numbers in the literature. **a**, From registered fundus-specific methods to a reproduced number; the first four stages follow the reproducibility levels of the registry. **b**, Share of method papers by split and by AUPR implementation (fixed thresholds, a grid of 11 or 33 operating points), and provenance of the baseline AUPR values on IDRiD and DDR in their comparison tables: produced by the reporting paper, identical to the method's own paper, identical to a value already carried by an earlier paper, or citing a source outside the ledger. **c**, mAUPR on IDRiD reported for one method in different papers (dots; bar, range; n, number of papers; methods reported by at least three papers) and the value after unified retraining (diamond). **d**, The same probability maps scored with the exact AUPR and with the 11-threshold implementation; the two shaded rows are the released M2MRF checkpoints, for which the authors' function on the float output is also shown. **e**, Rank of the retrained models under the two implementations.

0.036–0.038 for nnU-Net, M2MRF, HACDR-Net and HRDecoder, 0.084 for SAM2-UNet and 0.168 for FCT: the inflation is model-dependent and is largest for the models whose maps are least calibrated, so it reorders models that differ by less than it (Fig. 2d,e). Over the 19 retrained models, 10 changed rank under the 11-threshold implementation (Kendall $\tau = 0.88$ between the two orders).

Performance of the original models

Released checkpoints were scored with three numbers each: the value printed in the paper, the authors' evaluation function, and our protocol (Fig. 3a and Table 3). For M2MRF the authors' function reproduced the repository values to the last printed digit on both datasets (0.676 and 0.492), which shows that moving the code to a current framework did not change its output; the paper reports the mean of three runs (0.672 and 0.489). Under our protocol the same checkpoints reach 0.639 and 0.401, a gap of 0.036 and 0.091 that is almost entirely due to the threshold-0 term of the 11-threshold curve; restricting the evaluation to the field of view and keeping overlapping labels each change the third decimal only (Fig. 3b).

Rerunning the authors' recipe is the direct test of whether released code reproduces a reported number. A first rerun of M2MRF on IDRiD, with an effective batch of 1 instead of the authors' 4, reached 0.576 with the authors' metric and 0.493 with ours. With the authors' batch size of 4 and their 40,000 iterations, three seeds of the rerun reached 0.605–0.627 with the authors' metric (mean 0.615, s.d. 0.011) and 0.540–0.566 under our protocol (mean 0.552), against 0.676 and 0.639 for the released checkpoint and 0.672 for the three-run mean printed in the paper: the released code trains models that are 5–7 points below the released weights on every seed, a gap about five times the seed spread of the rerun itself and of the same architecture under the unified protocol below (s.d. 0.010). Further reruns of this kind (the DDR recipe of M2MRF, and the recipe of HACDR-Net, for which no weights are released) were cut when the scope was fixed and are marked as not rerun in Fig. 3c; the two extra IDRiD seeds were added when cards were free. The released HRDecoder checkpoints reached

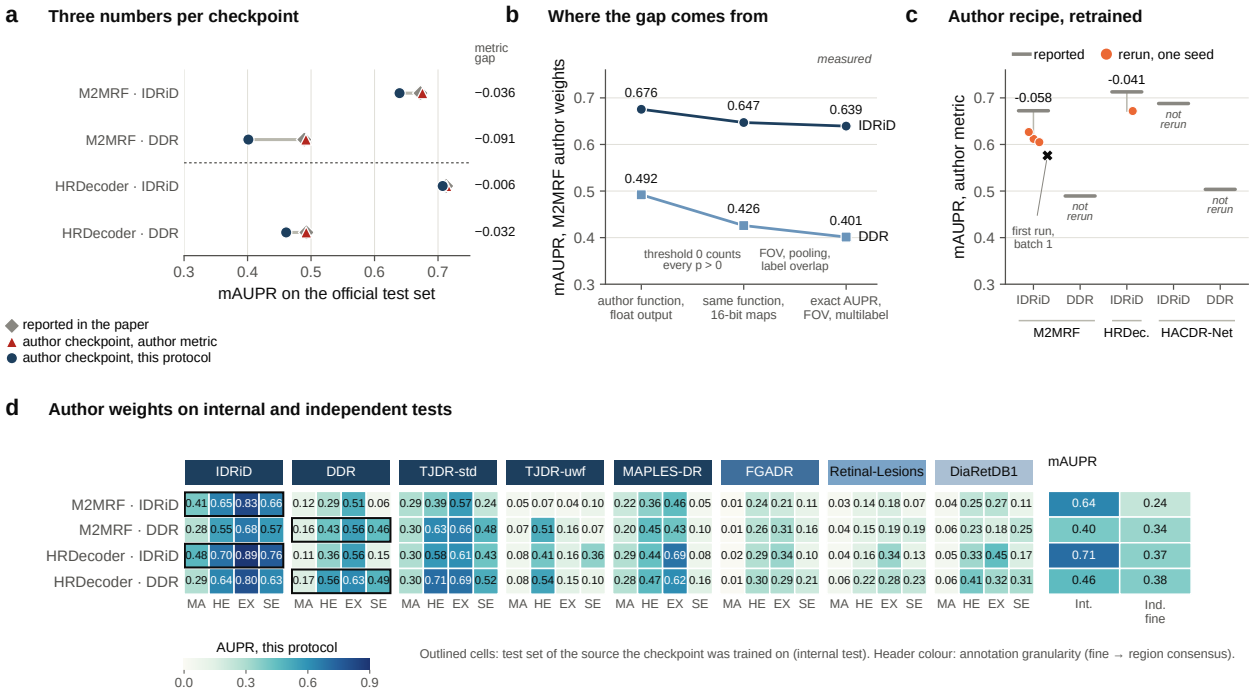


Fig. 3 Performance of the original models. a, mAUPR of each released checkpoint on the official test set of its training source: reported in the paper, authors' function on the checkpoint, and this study's protocol. The number on the right is the difference between the last two. **b**, Decomposition of that difference for the two M2MRF checkpoints. **c**, Retraining with the authors' recipe: reported value (line) and individual seeds (dots); the cross is the first rerun at an effective batch of 1. **d**, Per-class AUPR of the author weights on all eight test sets under this study's protocol. Outlined cells are internal tests.

0.707 on IDRiD and 0.461 on DDR under our protocol, as high as any model retrained below; they were trained by their authors on a canvas twice as large as the unified one and on three to five times as many pixels, so the two numbers are not comparable with the main table and are the reason for the backbone-controlled comparison in the next section. Rerun once with the authors' recipe (their geometry, augmentation, schedule and all 54 training images; one process in place of four), HRDecoder reached 0.672 with the authors' metric and 0.674 under our protocol on IDRiD, 0.04 and 0.03 below the released checkpoint (0.713 and 0.707). The authors' runner evaluates every 2,000 iterations on the test split and keeps the best checkpoint; the released file is the last iteration, and in our rerun that selection would have added 0.002, so it does not explain the distance between the two. Applied to the independent test sets as released, the checkpoints behaved like the retrained models (Fig. 3d): the DDR checkpoint scored higher on IDRiD (0.590) and on the standard-field TJDR set (0.554) than on its own test set (0.461), kept 0.382 on MAPLES-DR, and fell to 0.20–0.28 on the coarsely or regionally labelled sets and to 0.218 on the ultra-widefield images; the IDRiD checkpoint fell on every other source (0.165–0.478). The M2MRF checkpoints behaved the same way with lower values: the DDR checkpoint reached 0.519 on IDRiD and 0.518 on the standard-field TJDR set but 0.14–0.21 on the coarse, region-labelled and ultra-widefield sets, and the IDRiD checkpoint 0.244 on DDR, 0.372 on the standard-field TJDR set and 0.063 on the ultra-widefield images. WSRFNet, the third fundus-specific method with public inference code, releases no weights; they were requested from the authors and had not arrived, so no checkpoint of it is scored.

A unified protocol to retrain the models

Source recipes cannot serve as a common ground for retraining. The recipes of the general-purpose networks were written for natural images or abdominal organs; moved to IDRiD unchanged they differ about 67-fold in the number of training images seen, and those that resize the whole image to a small fixed input discard a large part of the small lesions (Fig. 7b). We therefore fixed everything that is not part of a method: data and splits, a canvas of 1,440 pixels across the field of view, the training amount counted in pixels, augmentation, selection of the checkpoint by validation mAUPR, inference geometry and evaluation. Each model keeps its loss, optimizer, learning rate, weight decay, schedule shape and effective batch (Methods and Table 8). The training amount was frozen from the validation curves of two models; no test metric was used. We retrained 19 models on IDRiD and on DDR: three seeds for U-Net, DeepLabv3, HRNet-W48, Swin-Unet, nnU-Net, M2MRF, HACDR-Net, SAMed and RETFound with UPerNet (FCT on IDRiD only), one seed for the other ten configurations (Supplementary Table 3); tables, the rank tests and the paired tests use the mean over the available seeds; where a result is quoted for seed 0 the text says so. A second track that retrains each model with its source recipe on the shared data was started for six runs and stopped for compute before any of them finished; it is left as future work, and the

Table 3 Original models: three numbers per released checkpoint, and retraining with the authors' recipe. a, Each checkpoint is scored on the official test set of its training source with the number printed in the paper, the authors' evaluation function (11 thresholds, whole image, overwritten labels) on the float output, the same function on the stored 16-bit maps, and this study's protocol (exact pooled AUPR inside the field of view, overlapping labels); under each per-class value is its 95% bootstrap interval over test images. **b**, The authors' pipeline rerun from scratch; last iteration, no checkpoint selection.

a Released checkpoints (author weights)

		mAUPR, author metric			AUPR, this protocol						
Model	**Source**	Paper	Function	Stored	**mAUPR**	MA	HE	EX	SE	**Δ metric**	**Indep. fine**
M2MRF	IDRiD	.672	.676	.647	.639	.408 .35–.42	.654 .57–.66	.832 .72–.85	.664 .44–.76	▼.036	.236
M2MRF	DDR	.489	.492	.426	.401	.165 .12–.20	.426 .28–.48	.558 .47–.61	.456 .29–.55	▼.091	.341
HRDecoder	IDRiD	.713	.713	.715	.707	.475 .44–.51	.699 .65–.74	.892 .81–.93	.763 .58–.87	▼.006	.369
HRDecoder	DDR	.493	.492	.492	.461	.172 .13–.22	.557 .51–.60	.629 .56–.70	.486 .33–.60	▼.032	.384

b Retrained with the authors' recipe (author-recipe reruns)

Model	**Source**	**Paper**	**Retrained, author metric**	**Seeds**	**Δ to paper**	**Retrained, this protocol**
M2MRF	IDRiD	.672	.615 ± .011	3	▼.058	.552 ± .013
first run, batch 1	IDRiD	.672	.576	1	▼.096	.493
HRDecoder	IDRiD	.713	.672	1	▼.041	.674

"Δ metric" is this protocol minus the authors' function on the same checkpoint. "Indep. fine" is the mean mAUPR on the three fine-grained independent test sets (TJDR standard-field, TJDR ultra-widefield and MAPLES-DR). The M2MRF paper reports the mean of three runs (67.24 on IDRiD, 48.94 on DDR), the repository the best run (67.55, 49.20). HRDecoder: the released checkpoints (repository table 71.34, 49.24) through the authors' test pipeline (resize to 2880 × 1920 with sliding 1920 windows on IDRiD, 2048 × 2048 whole image on DDR). The first M2MRF rerun used an effective batch of 1 instead of the authors' 4.

recipes themselves are in Table 8.

Performance of retrained models on internal tests

On the internal tests (Fig. 4a and Table 4), among the networks trained from their published initialisations the general-purpose HRNet-W48 reached the highest mAUPR on both sources, 0.700 ± 0.004 on IDRiD and 0.447 ± 0.009 on DDR (mean ± s.d. over three seeds), followed on IDRiD by Swin-Unet (0.688 ± 0.003) and the self-configuring nnU-Net (0.665 ± 0.008), and on DDR by nnU-Net (0.431 ± 0.003) and Swin-Unet (0.431 ± 0.001). The two fundus-specific networks retrained first did not lead under equal conditions: M2MRF reached 0.657 ± 0.010 and 0.369 ± 0.011, within 0.02 of the released checkpoint on IDRiD (0.639) and below it on DDR (0.401), and HACDR-Net 0.538 ± 0.031 and 0.252 ± 0.017 against 0.688 and 0.504 in its paper. The residual-encoder variant of nnU-Net did not improve on the default plan (0.631 and 0.428). The difference between the first two of these models, 0.012 on IDRiD and 0.016 on DDR, is of the order of the median seed standard deviation of the table (0.010 and 0.009), whereas the span of the table (0.41 on IDRiD, 0.28 on DDR) is thirty to forty times larger. The 95% bootstrap intervals over test images (seed 0; Supplementary Table 4) are wider still: their median width across models is 0.08–0.12 AUPR for microaneurysms, hemorrhages and hard exudates on IDRiD and 0.07–0.19 on DDR, and 0.33 and 0.24 for soft exudates, which appear in few test images. IDRiD has 27 test images, and its intervals are correspondingly wide; we use it to connect to the literature and base comparisons between architectures on DDR and on the independent tests.

Lesion classes differed more than models. The median AUPR over the 19 models was 0.87 for hard exudates and 0.44 for microaneurysms on IDRiD, and 0.60 and 0.15 on DDR; the best microaneurysm score on DDR was 0.19 (RETFound with a UPerNet head). Two architectures lost this class almost entirely: DeepLabv3, whose torchvision implementation decodes at an output stride of 8 with bilinear upsampling (microaneurysm AUPR 0.101 on IDRiD and 0.043 on DDR, hemorrhages and exudates normal), and FCT, whose recipe trains from scratch with a constant learning rate and did not converge within the budget (0.298 ± 0.084 on IDRiD, 0.176 on DDR; its three IDRiD seeds span 0.24–0.40). One failure was specific to a loss rather than to an architecture: the reference U-Net, trained on patches with a per-image Dice term smoothed by 1, drove its rare-class channels to zero on DDR (mAUPR 0.119, 0.119 and 0.103 over three seeds, with soft exudate AUPR 0.000), because 83% of DDR training patches contain no soft exudate and an empty patch rewards an all-zero output. Computing the Dice term over the batch, the first of the two recipe revisions made in this study (Methods), raised U-Net to 0.644 ± 0.015 on IDRiD and 0.371 ± 0.009 on DDR; this revision was decided from training and validation curves only and is disclosed as such. Two adapted foundation models went above HRNet-W48 on both sources on seed 0, and neither stayed clearly above it over three seeds. SAMed, a low-rank adaptation of SAM ViT-B trained on 512-pixel windows of the 1,440 canvas, reached 0.717 on IDRiD and 0.469 on DDR on seed 0 and 0.706 ± 0.015 and 0.460 ± 0.027 over three seeds (its DDR seeds span 0.430–0.482); RETFound with a UPerNet head 0.708 and 0.466 on seed 0 and 0.699

$\pm$ 0.009 and 0.450 $\pm$ 0.018 over three. Their seed-0 margins over the three-seed mean of HRNet-W48 on DDR (0.022 and 0.019) shrank to 0.013 and 0.003 over three seeds, within the seed scatter of the models themselves, and on IDRiD the three-seed margins (0.006 and -0.001) are within the width of the 27-image test set. SAM-Adapter followed (0.706 and 0.433). MedSAM, fully fine-tuned, lost both rare classes on DDR under its registered per-image Dice (microaneurysm AUPR 0.000, soft exudate 0.001; 0.265 overall), the third case of the mechanism described above; with the Dice term pooled over the batch under the rule registered for that case it reached 0.642 on IDRiD and 0.413 on DDR (microaneurysm 0.136, soft exudate 0.361), below SAMed, SAM-Adapter and RETFound with UPerNet on both sources (Methods; the collapsed runs are kept as the registered recipe). On DDR, RETFound with UPerNet gave the highest microaneurysm score of the table, 0.189 (0.14–0.24) (seed 0; 95% bootstrap interval in brackets), against 0.155 (0.12–0.20) for HRNet-W48 and 0.154 (0.11–0.17) for nnU-Net, and HRNet-W48 the highest soft-exudate score (0.550 (0.42–0.65)); the intervals of the leading models overlap on every class. SAM2-UNet, as registered, trained normally on IDRiD (0.636) but lost both rare classes on DDR (microaneurysm AUPR 0.023, soft exudate 0.022; 0.210 overall): its weighted IoU term is computed per image with smoothing 1, the mechanism that emptied the U-Net's rare-class channels, and under the rule registered for that case the term was pooled over the batch and the model retrained on both sources, to 0.664 and 0.360 (Methods; the collapsed run is kept as the registered recipe, Supplementary Table 6). Of the fundus-specific networks added from their code or papers, TC-Net (0.653 and 0.361) and PMCNet (0.652 and 0.312) were level with U-Net on IDRiD and below it on DDR, WSRFNet (0.582 and 0.234) was above only DeepLabv3 and FCT on IDRiD and above only FCT on DDR, and CLC-Net, re-implemented from its paper, reached 0.661 and 0.414, the highest DDR score of the fundus-specific family and above DeepLabv3 and U-Net. HRDecoder, M2MRF and HRNet-W48 share the HRNet-W48 backbone, so their three rows compare a general head with two lesion-specific decoders on one backbone. Under the unified protocol the general head led on both sources (0.700 and 0.447), M2MRF followed (0.657 and 0.369) and HRDecoder was last (0.647 and 0.296), a span of 0.05 on IDRiD and 0.15 on DDR; the released HRDecoder checkpoints, trained by their authors on a canvas twice as large and on three to five times as many pixels, score 0.707 and 0.461 under the same evaluation (Table 3). Within the budget and canvas of the unified protocol the lesion-specific decoders did not add to the backbone. Three further runs separate the explanations. The authors' recipe, rerun once, gave 0.674 on IDRiD under our protocol (previous section), 0.03 above the unified run and 0.03 below the released checkpoint. Doubling the training amount of the unified protocol at the 1,440-pixel canvas did not close the gap: 0.615 on IDRiD, below the single-budget run (0.647), and 0.304 on DDR against 0.296 (one seed each), whereas the same doubling raised the shared-backbone control HRNet-W48 on DDR from 0.447 to 0.469. The unified protocol at the authors' 2,880-pixel canvas and 1,920-pixel window, with the training amount counted in equivalent epochs at that canvas (the same 21,000 steps, four times the pixels per step), gave 0.627, below the 1,440-pixel run (0.647) although its validation score was higher (0.596 against 0.562 on the ten validation images); hard exudates gained (0.864 against 0.823) and soft exudates lost (0.615 against 0.728). Neither the canvas nor the training amount therefore accounts for the released checkpoint. What the authors' recipe adds beyond them, 0.03–0.06 on one seed, lies in what the protocol replaces: rotations of $\pm 45^\circ$ without photometric distortion, training on all 54 images with no validation split, and the last-step checkpoint (Supplementary Table 3). The adapted foundation models in the lower part of the table are described with the foundation-model results below.

Performance of retrained models on independent tests

Moving to a source that was never used for training did not by itself cost performance; the annotation tier and the imaging field did (Fig. 4b,c and Table 5; the 19 models on all eight test sets, seed means as in Table 4). The models trained on DDR averaged 0.369 mAUPR on the DDR test set and scored higher on the two other standard-field sources with pixel-level labels, 0.550 on IDRiD and 0.490 on the standard-field TJDR set (every model higher; two-sided Wilcoxon signed-rank $P < 0.001$), and the same on MAPLES-DR (0.372; 6 of 19 lower, $P = 0.595$). They lost a third to a half of their internal score wherever the labels or the geometry differ from the training source: 0.211 on FGADR (-43%) and 0.193 on Retinal-Lesions (-48%), whose lesions are drawn as coarse regions, 0.251 pixel-level mAUPR against the region-level consensus of DiaRetDB1 (-32%), and 0.173 on the 133° ultra-widefield TJDR images (-53%; every model lower on each of the four sets, $P < 0.001$). The models trained on the 44 IDRiD images started higher internally (mean 0.630) and ended lower on every other source, fine-grained or not: 0.286 on DDR, 0.424 on the standard-field TJDR set and 0.350 on MAPLES-DR (-33% to -55%), and 0.172–0.239 on the coarse, region-level and ultra-widefield sets (-62% to -73%; every model lower on every set, $P < 0.001$). The 0.26 advantage of IDRiD over DDR on the internal tests therefore reversed outside the training source: on the six independent sets the DDR-trained checkpoint of a model was better than its IDRiD-trained one in 74 of 114 pairs, in 13–15 of 19 on each of the five standard-field sets; only on the ultra-widefield images were the IDRiD-trained checkpoints usually ahead (4 of 19 DDR-trained ahead), consistent with the larger field and higher resolution of their training images. For both training sources the pixel-labelled standard-field sets were the least affected (a gain from DDR, the smallest loss from IDRiD) and the coarse, region-labelled and ultra-widefield sets the most. On the ultra-widefield images the inference scale mattered less than the field itself: resizing them to the 1,440-pixel canvas instead of predicting at native resolution with sliding windows, the comparison variant of the registered plan, moved the median mAUPR by +0.02 from both sources (13 and 14 of 19 models higher at the smaller scale, differences from -0.12 to +0.13), far from the 0.2–0.4 that separate this set from the standard-field ones (Extended Data Fig. 9b).

Table 4 Unified retraining: internal tests. Every model is trained on one source under the shared protocol and scored on that source's official test set. Per-class values are exact pooled AUPR inside the field of view, averaged over seeds; mAUPR is the mean of the four classes ± s.d. over seeds for the models trained with three seeds (no s.d. is printed for the single-seed rows; Supplementary Table 3 lists every run); Dice is pooled, at the per-class threshold chosen on the validation set. 95% bootstrap intervals over test images are given per class and model in Supplementary Table 4. Cell shading follows AUPR. **Bold**, best in column; underlined, second.

		Trained and tested on IDRiD (44 / 10 / 27)						**Trained and tested on DDR** (383 / 149 / 225)					
Model	**Params**	MA	HE	EX	SE	**mAUPR**	**Dice**	MA	HE	EX	SE	**mAUPR**	**Dice**
General segmentation													
U-Net	7.8	.403	.595	**.894**	.685	.644 ± .015	.622	.137	.405	.562	.380	.371 ± .009	.405
DeepLabv3	58.6	.101	.641	.739	.668	.537 ± .006	.515	.043	.460	.600	.405	.377 ± .005	.380
HRNet-W48	65.8	.449	.682	.884	**.784**	.700 ± .004	.627	.156	.500	.597	**.535**	.447 ± .009	.448
Swin-Unet	27.2	.470	.706	.894	.683	.688 ± .003	.640	.174	.455	**.654**	.439	.431 ± .001	.440
FCT	1.9	.081	.304	.680	.128	.298 ± .084	.330	.001	.163	.524	.014	.176	.197
Self-configuring													
nnU-Net	59.2	**.499**	.665	.872	.624	.665 ± .008	.656	.157	.503	.614	.449	.431 ± .003	.458
nnU-Net ResEnc-M	105.9	.449	.663	.878	.533	.631	.638	.147	.481	.653	.432	.428	**.463**
Fundus-specific													
M2MRF	70.3	.450	.628	.867	.681	.657 ± .010	.648	.148	.406	.534	.386	.369 ± .011	.407
HACDR-Net	24.6	.366	.463	.837	.484	.538 ± .031	.520	.105	.263	.487	.155	.252 ± .017	.289
HRDecoder	65.6	.410	.629	.823	.728	.647	.636	.124	.318	.496	.245	.296	.373
PMCNet	12.9	.418	.610	.883	.698	.652	.616	.157	.346	.482	.262	.312	.338
WSRFNet	9.7	.388	.588	.849	.502	.582	.641	.088	.292	.422	.135	.234	.341
CLC-Net^{C}	119.7	.437	.683	.889	.634	.661	.632	.152	.466	.612	.428	.414	.462
TC-Net^{C}	14.9	.355	.650	.854	.754	.653	.607	.112	.476	.607	.250	.361	.352
Foundation models, adapted													
SAMed	91.6	.439	.724	.889	.772	**.706** ± .015	.658	.172	**.559**	.638	.471	**.460** ± .027	.454
SAM-Adapter	641.3	.451	**.740**	.882	.750	.706	**.663**	.147	.538	.592	.454	.433	.457
SAM2-UNet	216.5	.439	.670	.871	.678	.664	.653	.153	.493	.518	.278	.360	.413
MedSAM (full FT)	93.7	.373	.665	.855	.674	.642	.609	.136	.529	.626	.361	.413	.432
RETFound + UPerNet	352.6	.480	.721	.889	.706	.699 ± .009	.650	**.193**	.453	.641	.516	.450 ± .018	.447

Params in millions. nnU-Net uses its own preprocessing and planning with the same splits, test images and evaluation. nnU-Net plans its network per dataset: the default configuration has 59.2 M parameters on both sources, the ResEnc-M preset 105.9 M under its IDRiD plan (printed) and 140.0 M under its DDR plan. ^{C}, no author code: re-implemented from the paper (Track C); the row shows what the published description yields under this protocol, not the authors' implementation. HRDecoder, M2MRF and HRNet-W48 share the HRNet-W48 backbone. H2Former is registered in tier 1 but deferred: at its 960-pixel input one training step exceeds the 40 GB of the cards used here.

Internal rank predicted independent rank where the labels were drawn the same way. For DDR-trained models the Spearman correlation between internal mAUPR and the score on IDRiD, the standard-field TJDR set and MAPLES-DR was 0.92, 0.88 and 0.82: HRNet-W48, third internally, was first on IDRiD (0.643) and on the standard-field TJDR set (0.616), SAMed, first internally, on MAPLES-DR (0.470) and FGADR (0.288; $\rho = 0.79$), and SAMed also led the mean of the three fine-grained independent sets (0.426; HRNet-W48 0.412). On Retinal-Lesions and DiaRetDB1 the correlation was 0.76 and 0.78 but the first place went to DeepLabv3 (0.282 and 0.348), tenth internally, whose output stride of 8 blurs its maps into region-sized blobs, and on the ultra-widefield images the correlation vanished ($\rho = 0.01$) and the fully fine-tuned MedSAM, ninth internally, was first (0.243). For IDRiD-trained models the correlation was 0.75 on DDR and 0.71 on MAPLES-DR and 0.35–0.53 elsewhere; SAMed led DDR, the standard-field TJDR set and FGADR (0.368, 0.534, 0.280) and the fine-grained mean (0.394), whereas TC-Net, tenth internally, led MAPLES-DR (0.412), DeepLabv3, eighteenth, led Retinal-Lesions and DiaRetDB1 (0.260 and 0.311), and MedSAM, fourteenth, the ultra-widefield images (0.295). RETFound with UPerNet, second internally on DDR, was the weakest model after the non-converged FCT on the ultra-widefield images from DDR (0.111): its 224-pixel windows see 1/19 of the 4,256-pixel canvas at a time.

Internal and independent performance were correlated across models (Spearman $\rho = 0.80$ for DDR-trained and 0.65 for IDRiD-trained models, internal mAUPR against the mean of the three fine-grained independent sets), and the model that led the fine-grained independent sets, SAMed (0.43 from DDR, 0.39 from IDRiD), was the same model as on the internal test. Averaged by family over both sources, the adapted foundation models were ahead on internal (0.55) and on fine-grained independent tests (0.38), the self-configuring baseline followed (0.54 and 0.36), and the fundus-specific (0.47 and 0.32) and general networks (0.47 and 0.30) were level with each other (Fig. 4d).

Over the eight test-set blocks the Friedman test rejected equal ranks ($\chi^2 = 97.9$, $P < 0.001$, 19 models). SAMed had the best average rank (1.63), followed by HRNet-W48 (4.13) and Swin-Unet (5.25), and FCT the worst (17.88); with eight blocks the Nemenyi critical difference is 9.90 ranks, so 11 of the 19 models were not separable from the first (Fig. 4e). The ranking published for the 10 models that report an IDRiD mAUPR did not agree with the ranking after retraining (Kendall τ = -0.11, $P = 0.73$; Fig. 4f). A threefold change of the learning rate, tested on DDR for one general and one fundus-specific network and judged on validation mAUPR only, changed the order of HRNet-W48 and M2MRF at one third of the learning

rate (0.524 against 0.553) and not at three times it (0.630 against 0.525; main-table rates 0.617 and 0.558), so the main-table learning rates were kept (Extended Data Fig. 7). Averaging the probability maps of the retrained models, in compositions fixed before any ensemble was scored (Methods), gave the highest scores of the study on every test set from DDR and on all but one from IDRiD (Table 5, lower block; seed-0 members; on FGADR and Retinal-Lesions the two compositions that contain HACDR-Net were computed without it, because its maps for those two sets could not be regenerated, and are marked in the table). From DDR, the mean of all 18 members (the models with a seed-0 run when the compositions were fixed; MedSAM is not a member) reached 0.505 on the DDR test set and the mean of the five best by validation mAUPR 0.511, against 0.469 for the best member; the same two ensembles reached 0.677 and 0.680 on IDRiD (best member 0.645), 0.637 and 0.652 on the standard-field TJDR set (0.624), 0.496 and 0.506 on MAPLES-DR (0.466) and 0.266 and 0.242 on the ultra-widefield images (0.208; the retrained MedSAM, which is not a member, reached 0.243 there). From IDRiD the mean of all members reached 0.764 on its own test set (best member 0.717), 0.594 on the standard-field TJDR set (0.559) and 0.400 on DDR (0.380). On the coarsely and region-labelled sets the gain was smaller: the five best DDR-trained members averaged reached 0.298 on FGADR and 0.290 on Retinal-Lesions against 0.290 and 0.278 for the best member, the mean of the remaining 17 members 0.291 and 0.278, no better than the best member, and the mean of all members 0.335 on DiaRetDB1 against 0.325. The gain over the best member was thus 0.02–0.06 mAUPR on the fine-grained test sets and −0.01 to 0.02 on the others (from IDRiD, DeepLabv3 alone was higher than every ensemble on Retinal-Lesions), and the four adapted foundation models averaged by themselves matched or exceeded the ensemble of all models (0.509 on DDR, 0.676 on IDRiD, 0.657 on the standard-field TJDR set, 0.302 on FGADR, 0.294 on Retinal-Lesions), whereas the seven fundus-specific networks averaged by themselves gave the weakest family ensemble (0.444 on DDR). Ensembles are a reference for users and do not enter the rank tests above. A reference model that follows these findings, the two members with the highest DDR validation score (RETFound with UPerNet and SAMed) each trained once on the three fine-grained sources merged and then averaged (Methods), is released with the benchmark (`bench.reference`: weights, thresholds chosen on the DDR validation split, and an entry point that writes probability maps and masks for new images). Trained on the merged sources, the two members reached 0.477 (SAMed) and 0.451 (RETFound with UPerNet) on the DDR test set, against 0.469 for the best single DDR-trained model (SAMed, seed 0); the averaged model's scores on all eight test sets, three of them in-source, are given in Supplementary Table 11 and are not ranked.

Scores also depend on the disease stage of the test images, because the pooled AUPR follows the number of lesion pixels available. On the four test sets that carry a DR grade per image, every score was recomputed within grades 0–1 and within grades 2–4 with the thresholds of the full test (Methods; Supplementary Table 12). On the DDR test set the median mAUPR of the DDR-trained models (seed 0) was 0.21 on the 20 images of grades 0–1 and 0.36 on the 203 images of grades 2–4, on FGADR 0.04 and 0.22 (94 and 459 images) and on Retinal-Lesions 0.07 and 0.17 (150 and 328); the image-level AUC on DDR was 0.54 within grades 0–1, where few images carry a lesion of the class (defined for 12 of the 19 models), and 0.75 within grades 2–4. MAPLES-DR is graded on the Canadian teleophthalmology scale, under which most of its images are R1 (mild) although they were drawn from MESSIDOR grades 1–3[42], so 56 of its 60 test images fall in grades 0–1 (median 0.33) and only 4 in grades 2–4 (0.40), too few for a comparison. The grade mix of a test set therefore sets the level of every model's score, and the comparisons of this study hold within a test set and not across sets.

DiaRetDB1 gives the comparison a human reference. Each of its four experts, scored against the consensus of the other three with the same protocol, reached a pixel-level mAUPR of 0.15–0.28 (median 0.21) and an image-level AUC of 0.87–0.93 (Table 5, bottom block, and Supplementary Table 13; a four-level confidence map yields a four-point precision–recall curve, so the pixel-level value is a lower bound). The retrained models scored against the four-expert consensus (seed 0) spanned 0.09–0.32 pixel-level mAUPR (median 0.28 from DDR, 0.25 from IDRiD) and 0.71–0.94 image-level AUC (median 0.84 and 0.83), so on pixel-level agreement with the consensus all but 3 of the 38 model–source pairs lie within or above the band of the experts, whereas on image-level detection only 7 (DDR-trained) and 5 (IDRiD-trained) of 19 reach the experts' lowest value of 0.87; the experts' own microaneurysm agreement was the lowest of the four classes (AUPR 0.001–0.022 against 0.039 for the median model), which says as much about the consensus rule as about the models.

Foundation models under four kinds of use

Results for segmentation foundation models are often quoted without saying how the model was used. We separated four kinds of use and report them side by side without ranking one against another (Fig. 5a and Table 6): zero-shot use without any label; prompts derived from the ground truth, which is an upper bound and not a usable setting; parameter-efficient fine-tuning; and an encoder with a trained decoder or full fine-tuning.

Zero-shot use did not segment DR lesions. The automatic masks of SAM and SAM 2, with the density and area cap chosen on the validation pool, reached an any-lesion AUPR of 0.07 and 0.08 on IDRiD and 0.03 on DDR, against 0.78 for nnU-Net's maps on the same metric, and even when each mask was given the class it overlaps most in the ground truth, an upper bound that reads the test labels, the mAUPR stayed at 0.03. Text prompts to SAM 3 found two of the four classes: with the wording chosen on the validation pool, hard exudates reached an AUPR of 0.41 on IDRiD and hemorrhages 0.29 (0.24 and 0.23 on DDR), soft exudates 0.21 on IDRiD and 0.02 on DDR, and microaneurysms 0.01 under every wording; the wording mattered as much as the class, from 0.03 to 0.41 for hard exudates and from 0.00 ("cotton wool spot") to 0.21 ("white fluffy patch") for soft exudates (Fig. 5c). Prompts derived from the ground truth tell where the gap lies. With a box

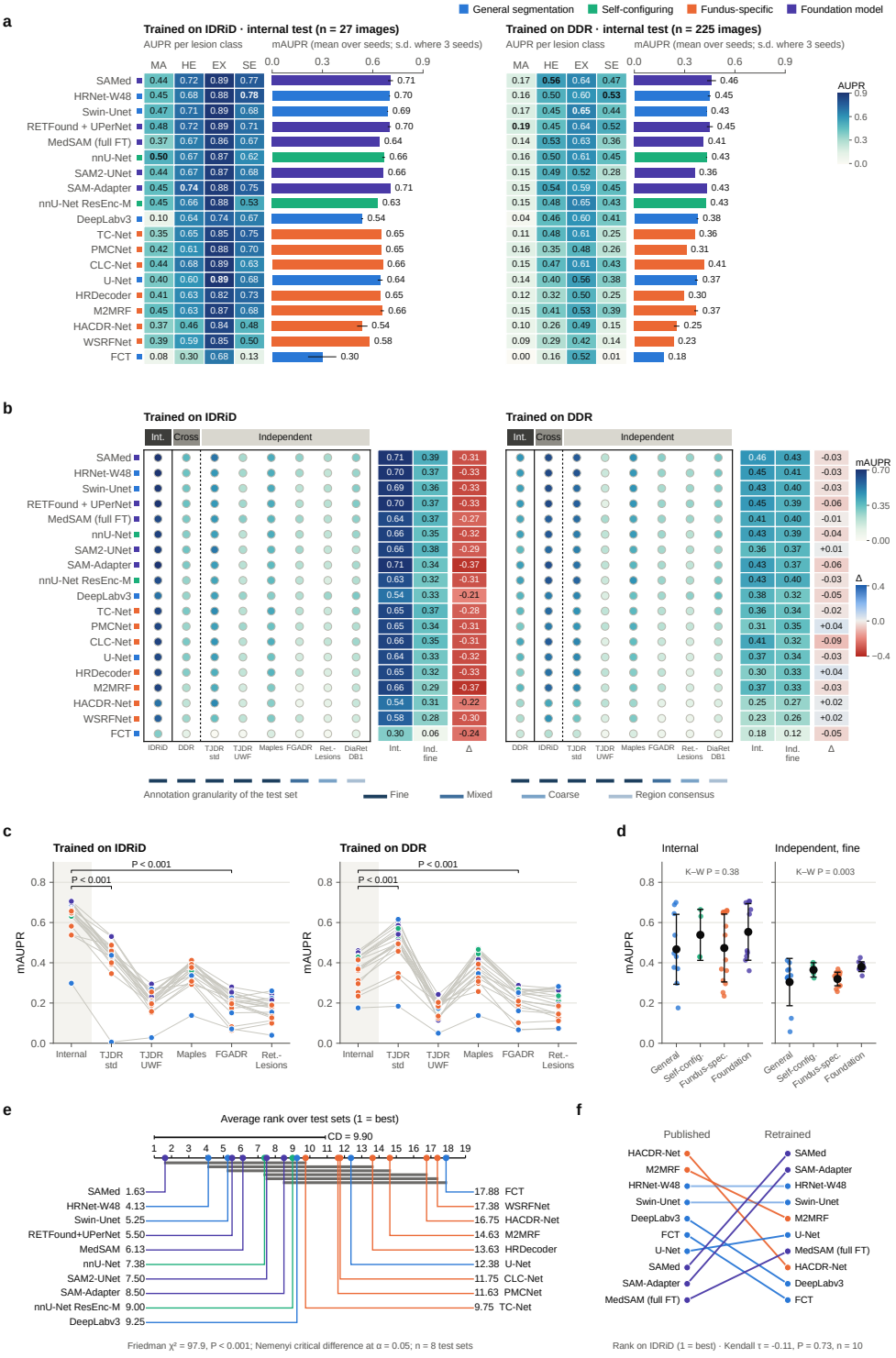


Fig. 4 Performance of retrained models under the unified protocol. a, Internal tests: per-class AUPR (mean over seeds) and mAUPR (bar; error bar, s.d. over seeds where three were run). Models are ordered by their average rank in **e**; the square gives the family. **b**, mAUPR of every model on the eight test sets, by training source. Int., internal test; Cross, test set of the other training source; the bar under each column gives the annotation granularity. Columns on the right: internal test, mean of the three fine-grained independent test sets, and their difference. **c**, The same models on the internal test and on four independent test sets; each line is one model. P values, two-sided Wilcoxon signed-rank tests across models with Holm correction ($n = 19$). **d**, mAUPR by family on internal and on fine-grained independent tests; coloured dots, one model and training source; black, mean ± s.d.; K–W, Kruskal–Wallis test. **e**, Critical-difference diagram over the eight test sets; bars join models whose average ranks differ by less than the Nemenyi critical difference. **f**, Rank on IDRiD as published and after retraining.

around every ground-truth component, an upper bound that reads the test labels, SAM 2 reached 0.85 mAUPR on IDRiD, but an IDRiD image carries on average 44 microaneurysm and 142 hard exudate components, and with 1, 5 and 20 boxes per image and class the mAUPR was 0.16, 0.35 and 0.55 (Fig. 5b); one positive point per component, the cheapest prompt, gave 0.12 on IDRiD and 0.02 on DDR, with only soft exudates (0.41) above 0.03. Given every box, the supervised models' own maps cut to the union of the same boxes scored 0.86 (nnU-Net) and 0.81 (M2MRF) on IDRiD and 0.71 and 0.67 on DDR, against 0.86 and 0.77 for SAM ViT-B, 0.85 and 0.76 for SAM 2 and 0.66 and 0.65 for MedSAM: inside a box the general promptable models delineated the lesion as well as the best supervised model on IDRiD and better on DDR, and the medically fine-tuned MedSAM worse than both. The gap of the promptable models therefore lies in finding lesions, not in drawing them.

Among the trained settings, SAMed was the best parameter-efficient adaptation (0.460 ± 0.027 on DDR over three seeds, against 0.431 for nnU-Net), and full fine-tuning of MedSAM (0.413) was below every parameter-efficient adaptation of SAM except SAM2-UNet (Table 6). The pretraining of an encoder was then tested under one UPerNet head, one recipe and one training amount on DDR, in two pairs fixed in advance: an encoder pretrained on retinal images against its own starting point, and the same contrast under a second pretraining objective, so that the effect of retinal pretraining does not depend on one objective (Fig. 5d,e). In the first pair, RETFound's masked-autoencoder encoder (0.47 internal, 0.59 on the standard-field TJDR set) was below the ImageNet masked autoencoder it was initialised from (0.52 and 0.63), by 0.053 internally and 0.045 on the standard-field TJDR set, and below it on the ultra-widefield images (0.11 against 0.13) and on MAPLES-DR (0.46 against 0.48) as well. In the second pair, RETFound's DINOv2 encoder reached 0.49 internally and 0.62 on the standard-field TJDR set, against 0.51 and 0.62 for DINOv2 itself under the same head: -0.020 internally, and within 0.015 on each of the three independent fine-grained test sets (0.15 against 0.14 on the ultra-widefield images, 0.50 against 0.49 on MAPLES-DR; mean over the three 0.42 against 0.41). Under one head, one recipe and one seed per encoder, retinal pretraining therefore added nothing to either of its starting points, under either objective; three of the four encoders, both natural-image encoders and RETFound's DINOv2 encoder, exceeded every row of Table 4 on DDR, and the fourth is the main table's RETFound row. The three control encoders were trained after the 19 rows of the main table had been registered, so they are reported here and in Supplementary Table 10 rather than as ranked rows; with one seed each, the ImageNet masked autoencoder under the UPerNet head nevertheless gave the highest single-model DDR score of the study (0.52, against 0.469 for the best ranked model on seed 0), 0.05 above the best ranked foundation model and 0.08 above the best convolutional network, a margin larger than the seed scatter of any ranked model. Trained on IDRiD, where the decision to add the three control cells was taken after the DDR result of the first pair had been read, the two natural-image encoders reached 0.73 (masked autoencoder) and 0.74 (DINOv2) on the IDRiD test set, again above every row of Table 4 (best ranked model 0.717 on seed 0), against 0.71 for RETFound's masked-autoencoder encoder and 0.71 for its DINOv2 encoder under the same head, and 0.41 and 0.41 against 0.37 and 0.40 on the mean of the three independent fine-grained sets; both pair differences had the same sign as on DDR (-0.021 and -0.028 internally; Supplementary Table 10).

Effects of annotation granularity and training source

The sources differ in how lesions were drawn before they differ in anything a model does (Fig. 6a; Methods). Counted as connected components of the prepared masks over every image of each source, an IDRiD image carries on average 43 microaneurysm and 144 hard-exudate components whose median diameter is 0.5% of the field-of-view width, and a DDR image 14 and 31 at 0.4%; MAPLES-DR, drawn pixel by pixel by retinologists, has 27 microaneurysms and 16 hard exudates per image at 0.8% and 1.0%. The two TJDR sets carry few components per image (3–7 per class) of the same small size (0.4–1.8%), so their difference to IDRiD and DDR is in how many lesions the images carry, not in how they were outlined. In the coarse and region-consensus sources the components are few and large: a Retinal-Lesions image carries 2 hard-exudate and 8 hemorrhage components with median diameters of 4.2% and 3.5% of the field, and a DiaRetDB1 image 1.5 and 4.5 at 4.1% and 2.9%, eight to nine times the hard-exudate diameter of IDRiD and DDR and two to four times their hemorrhage diameter, which for the same retina means that several lesions and the tissue between them were enclosed by one outline. FGADR, drawn as a mixture of single lesions and regions, sits between: 29 hard-exudate and 21 hemorrhage components per image, as many as DDR's, at 0.8% and 1.2%, somewhat larger than DDR's and a fifth to a third of Retinal-Lesions'.

To separate the effect of the training source from that of the model, the five representative models (U-Net, nnU-Net, M2MRF, SAM2-UNet and RETFound with UPerNet) were also trained on the standard-field TJDR images and on FGADR, at the pixel budget of DDR and with one seed, which together with the main table gives a matrix of four training sources by eight test sets (Fig. 6b,d; for the TJDR- and FGADR-trained models, whose sources fix no single field of view, the ultra-widefield set is predicted at the field-of-view scale, Methods). Every source served its own test set best: in-source training gave 0.670 mAUPR on IDRiD, 0.403 on DDR, 0.571 on the standard-field TJDR set and 0.411 on FGADR, against 0.390–0.584, 0.263–0.301, 0.450–0.523 and 0.173–0.232 from the other three sources (Fig. 6c). Training elsewhere cost 0.16 mAUPR on average over the twelve off-diagonal cells and up to 0.28 (TJDR-trained models on IDRiD). Which foreign source was best depended on the labels of the test set rather than on the number of training images: DDR, the largest fine-grained source (383 images), was the best foreign source on every standard-field pixel-level test set (0.584 on IDRiD, 0.523 on the standard-field TJDR set, 0.391 on MAPLES-DR), whereas FGADR, three times larger (1,096

images) but drawn as mixed regions, was the best foreign source only on the two sets with coarse or region labels (0.232 on Retinal-Lesions and 0.270 on DiaRetDB1, against 0.201 and 0.259 from DDR) and on the ultra-widefield images (0.247 against 0.173 from DDR; its cells are predicted at the field-of-view scale, Methods), and lost 0.15 on average on the three fine-grained sources relative to in-source training; the standard-field TJDR set, the smallest source (172 images), transferred worst (0.390 on IDRiD, 0.264 on DDR).

We then added mixed or coarse labels to fine-label training while keeping the number of patch pixels fixed, for U-Net and M2MRF with one seed; differences smaller than twice the median seed standard deviation of the main table (0.018 on DDR) are reported without interpretation (Fig. 6e and Table 7). On the DDR test set, adding the 1,096 FGADR images to DDR changed mAUPR by −0.01 for U-Net (0.378 to 0.373) and −0.06 for M2MRF (0.380 to 0.320); adding only 383 FGADR images, as many as DDR has, by −0.01 and −0.04 (0.369 and 0.336), so the loss is not a matter of the fine images being outnumbered; adding Retinal-Lesions by −0.04 and −0.09 (0.341 and 0.286). On the IDRiD test set the same three training sets moved U-Net by +0.05, +0.03 and −0.03 (from 0.547) and M2MRF by −0.02, −0.02 and −0.06 (from 0.571), and on the standard-field TJDR and MAPLES-DR sets U-Net by −0.05 to +0.03 and M2MRF by −0.18 to −0.01. Mixed labels added to fine ones were therefore at best neutral and at worst costly, and coarse labels cost M2MRF 0.06–0.18 on every standard-field fine-grained test set. Trained on Retinal-Lesions alone, the two models scored 0.504 and 0.469 on its own test split but 0.164 and 0.104 on DDR and 0.248 and 0.167 on IDRiD; trained on FGADR alone, 0.364 and 0.437 on IDRiD.

A revision of the labels themselves, as opposed to a change of source, moved the scores far less. A refined annotation of the 27 IDRiD test images was released in 2026 (Refined IDRiD; one class per pixel, 1,024-pixel crops of the retinal region). We located each crop in the original image (every crop is the retinal bounding box padded to a square and resized; normalised cross-correlation with the original $\geq$ 0.9988 for all 27), resampled the official labels through the same geometry, and scored the test maps of the 19 IDRiD-trained models of Table 4 against both label sets without retraining or re-inference. The revision changed 13 of the 27 images for microaneurysms and hemorrhages, 14 for hard exudates and 6 of 14 for soft exudates, moving 5.7%, 1.6%, 4.0% and 0.7% of the labelled pixels (pooled intersection over union 0.94, 0.98, 0.96 and 0.99) and no component count except hard exudates (3,895 to 3,882). Under the same protocol the mAUPR of every model changed by between -0.002 and 0.000 (microaneurysm AUPR by -0.002 on average, the other classes by less than 0.002), and the order of the 19 models was unchanged (Kendall τ = 1.00). The 1,024-pixel re-scoring itself stayed within 0.006 mAUPR of the main-table test at the 1,440-pixel canvas (τ = 0.95). What separates the sources in Fig. 6 is therefore how lesions were defined and outlined, which a correction of individual masks within one style does not touch (Supplementary Table 9).

Effects of lesion size, input resolution and training amount

Lesion size is the property that separates the four classes most clearly (Fig. 7a). On the training images the median diameter of a microaneurysm is 0.5% of the field-of-view diameter and that of a hemorrhage 1.1%; we froze these two values, computed on training images only, as the boundaries between small, medium and large lesions before the main-table training. Resizing a whole image to a small input removes lesions from the training target itself: at 224 pixels 16.0% of the microaneurysms and 25.8% of the hard exudates of the IDRiD training set no longer cover a single pixel, and on DDR 34.3% and 35.3%; at 384 pixels the DDR figures are 11.5% and 16.9%, and on the 1,440-pixel canvas of the unified protocol essentially none are lost (Fig. 7b).

Lesion-level recall followed size for every model (Fig. 7c). On the DDR test set the median recall over the 19 models (seed 0, threshold chosen on the validation set) was 0.26 for small, 0.49 for medium and 0.66 for large lesions; the best models detected 0.30 of the small ones (Swin-Unet and SAM2-UNet, nnU-Net ResEnc-M 0.29) and DeepLabv3, decoding at an output stride of 8, 0.07. By class, the median recall of small lesions was 0.46 for microaneurysms, 0.34 for hard exudates and 0.15 for hemorrhages, and the median lesion-level precision 0.29, 0.73 and 0.65. Input resolution acted mainly through this channel. Shrinking the canvas from 1,440 to 512 pixels at an equal pixel budget lowered mAUPR on DDR from 0.380 to 0.358 for M2MRF and from 0.466 to 0.400 for RETFound with UPerNet (one seed), the recall of small lesions from 0.27 to 0.18 and from 0.28 to 0.11, and the microaneurysm AUPR from 0.148 to 0.114 and from 0.189 to 0.092 (Fig. 7d). Patches at native resolution, where a 960-pixel patch covers a smaller part of the retina and the fixed pixel budget buys fewer passes over each image, raised the small-lesion recall of M2MRF to 0.33 and of RETFound to 0.31, but lowered mAUPR to 0.312 and 0.449: for RETFound the microaneurysm AUPR rose (0.189 to 0.196) while the hemorrhage and soft-exudate AUPR fell (0.491 to 0.460 and 0.546 to 0.513), so the smallest class gained from native resolution and the two largest lost.

More training images helped, with diminishing returns within the size of DDR. With 10%, 50% and 100% of the DDR training set (38, 192 and 383 images; equal number of patch pixels, so each image is seen ten, two and one times as often), mAUPR on the DDR test set was 0.321, 0.373 and 0.378 for U-Net and 0.307, 0.342 and 0.380 for M2MRF, and on IDRiD 0.492, 0.540 and 0.547 and 0.473, 0.549 and 0.571 (Fig. 7e). Merging the three fine-grained sources IDRiD, DDR and standard-field TJDR (599 images) at the same budget gave 0.367 and 0.368 on DDR, within 0.01 of DDR alone, and raised the scores on the sets outside the training sources by 0.01–0.04 (MAPLES-DR 0.372 and 0.349 against 0.329 and 0.324; FGADR 0.192 and 0.224 against 0.170 and 0.212; Retinal-Lesions 0.177 and 0.153 against 0.142 and 0.150), the gain of

more fine-grained images rather than of in-source data. A source merged into the training set raised the score on its own test set by more, 0.05–0.07 (standard-field TJDR 0.562 and 0.565 against 0.494 and 0.506; IDRiD 0.607 and 0.619 against 0.547 and 0.571), which is the in-source gain of the matrix above and not generalization; these two tests are in-source for the merged run and are marked as such in Fig. 7e. Exchanging the loss of U-Net with the architecture fixed moved mAUPR on DDR between 0.357 (cross-entropy alone) and 0.378 (cross-entropy with batch Dice, the main table); Dice alone gave 0.237, the per-image Dice of the registered recipe 0.119, the collapse described above (Extended Data Fig. 8).

Computational cost

All models were trained on single 40 GB cards, one task per card. Because the training amount is fixed in pixels, the cost of a model under the unified protocol reflects its throughput and its patch size: one seed on IDRiD took from 1.5 GPU hours (HRDecoder and U-Net) to 20 (FCT), and a DDR seed 3.9 times as long (median ratio), the slowest being the models that process small patches with sliding windows (FCT, 72 hours on DDR; SAM-Adapter 26, HACDR-Net 28 and CLC-Net 39 hours). nnU-Net follows its own schedule of 1,000 epochs rather than the pixel budget (7 hours per source, 11–13 for the residual-encoder plan). The two models that led the tables were not the costliest: SAMed took 2.1 and 7.4 hours and RETFound with UPerNet 6.1 and 24 hours per seed. Accuracy was not related to cost or to parameter count (Extended Data Fig. 10 and Supplementary Table 8).

Table 5 Unified retraining: independent tests. The checkpoints of Table 4 are applied without any adaptation to six test sets from sources never used for training. Values are mAUPR averaged over the seeds of Table 4; the DiaRetDB1 column is the official image-level ROC AUC on DiaRetDB1, averaged over the four lesion types, because its region-consensus ground truth biases pixel precision. Δ is the mean of the three fine-grained independent test sets (TJDR standard-field, TJDR ultra-widefield and MAPLES-DR) minus the internal test. Average rank is over the eight test sets of Fig. 4e (Friedman χ^2 = 97.9; Nemenyi critical difference 9.90). Per-class values with 95% bootstrap intervals are in Supplementary Table 4.

	Trained on IDRiD								Trained on DDR								
	TJDR								TJDR								Avg.
Model	Int.	Std	UWF	Maples	FGADR	Ret.-L.	DiaRet	Δ	Int.	Std	UWF	Maples	FGADR	Ret.-L.	DiaRet	Δ	**rank**
General segmentation																	
U-Net	.64	.44	.20	.34	.15	.15	.85	−.32	.37	.49	.17	.35	.16	.15	.84	−.03	12.38
DeepLabv3	.54	.43	.27	.29	.23	**.26**	.87	−.21	.38	.50	.13	.34	.25	**.28**	.90	−.05	9.25
HRNet-W48	.70	.45	.25	.40	.25	.19	.86	−.33	.45	**.62**	.17	.45	.27	.25	.87	−.03	4.13
Swin-Unet	.69	.47	.22	.40	.20	.22	**.91**	−.33	.43	.56	.18	.46	.24	.26	.92	−.03	5.25
FCT	.30	.01	.03	.14	.07	.04	.83	−.24	.18	.18	.05	.14	.07	.07	.82	−.05	17.88
Self-configuring																	
nnU-Net	.66	.43	.20	.40	.24	.19	.83	−.32	.43	.52	.20	.44	.26	.21	.82	−.04	7.38
nnU-Net ResEnc-M	.63	.42	.19	.36	.20	.14	.82	−.31	.43	.57	.17	.47	.27	.24	**.94**	−.03	9.00
Fundus-specific																	
M2MRF	.66	.40	.15	.31	.08	.11	.73	−.37	.37	.49	.18	.32	.21	.15	.78	−.03	14.63
HACDR-Net	.54	.49	.16	.29	.06	.10	.82	−.22	.25	.35	.20	.26	.10	.11	.74	+.02	16.75
HRDecoder	.65	.39	.19	.38	.20	.13	.80	−.33	.30	.50	.20	.30	.19	.13	.82	+.04	13.63
PMCNet	.65	.43	.26	.34	.21	.18	.81	−.31	.31	.46	.20	.39	.21	.19	.80	+.04	11.63
WSRFNet	.58	.35	.20	.31	.20	.13	.72	−.30	.23	.33	.14	.31	.07	.07	.81	+.02	17.38
CLC-Net	.66	.46	.21	.39	.17	.19	.79	−.31	.41	.50	.13	.35	.18	.15	.81	−.09	11.75
TC-Net	.65	.48	.21	**.41**	.21	.18	.88	−.28	.36	.47	.19	.36	.21	.18	.86	−.02	9.75
Foundation models, adapted																	
SAMed	**.71**	**.53**	.25	.40	**.28**	.24	.88	−.31	**.46**	.60	.21	**.47**	**.29**	.27	.89	−.03	**1.63**
SAM-Adapter	.71	.44	.21	.36	.22	.19	.71	−.37	.43	.54	.20	.37	.25	.19	.76	−.06	8.50
SAM2-UNet	.66	.53	.23	.37	.25	.19	.79	−.29	.36	.51	.20	.40	.25	.23	.84	+.01	7.50
MedSAM (full FT)	.64	.45	**.29**	.37	.26	.21	.84	−.27	.41	.55	**.24**	.42	.27	.26	.90	−.01	6.13
RETFound + UPerNet	.70	.45	.25	.40	.23	.21	.85	−.33	.45	.59	.11	.46	.27	.27	.88	−.06	5.50
Ensembles of the rows above (pixelwise mean of the probability maps; members chosen on validation data)																	
E-top3	.74	.56	.28	.44	.27	.25	.88	−.32	.51	.66	.23	.50	.30	.29	.89	−.05	–
E-top5	.75	.60	.29	.45	.29	.25	.88	−.31	.51	.65	.24	.51	.30	.29	.89	−.04	–
E-1perfam	.74	.56	.28	.45	.26	.24	.87	−.31	.49	.63	.23	.49	.28	.28	.89	−.05	–
E-all	.76	.59	.31	.46	.28^{P}	.22^{P}	.91	−.31	.50	.64	.27	.50	.29^{P}	.28^{P}	.87	−.04	–
E-fam. general	.73	.51	.28	.41	.24	.19	.88	−.33	.48	.61	.23	.47	.27	.27	.87	−.05	–
E-fam. auto	.69	.48	.23	.41	.24	.18	.81	−.32	.46	.57	.21	.47	.27	.23	.88	−.04	–
E-fam. fundus	.74	.57	.29	.44	.25^{P}	.21^{P}	.92	−.30	.44	.57	.25	.45	.25^{P}	.22^{P}	.85	−.02	–
E-fam. foundation	.76	.61	.30	.44	.29	.25	.86	−.31	.51	.66	.25	.49	.30	.29	.89	−.04	–
Human reference: one DiaRetDB1 expert against the consensus of the other three (leave-one-expert-out; no training source, same value under both blocks)																	
Expert 01	–	–	–	–	–	–	.87	–	–	–	–	–	–	–	.87	–	–
Expert 02	–	–	–	–	–	–	.93	–	–	–	–	–	–	–	.93	–	–
Expert 03	–	–	–	–	–	–	.93	–	–	–	–	–	–	–	.93	–	–
Expert 04	–	–	–	–	–	–	.91	–	–	–	–	–	–	–	.91	–	–
Expert range	–	–	–	–	–	–	.87–.93	–	–	–	–	–	–	–	.87–.93	–	–

Granularity of the test set: fine mixed coarse region consensus. Int., internal test (official test set of the training source). Std, standard field (35–50°); UWF, ultra-widefield (133°), predicted at native resolution with sliding windows; Maples, MAPLES-DR (official test split, 60 images). Retinal-Lesions excludes its grey-level 127 regions. Best and second-best are marked among the measured rows of a column. ^{P}, ensemble computed without a registered member whose maps could not be produced on that set (named in the run's record).

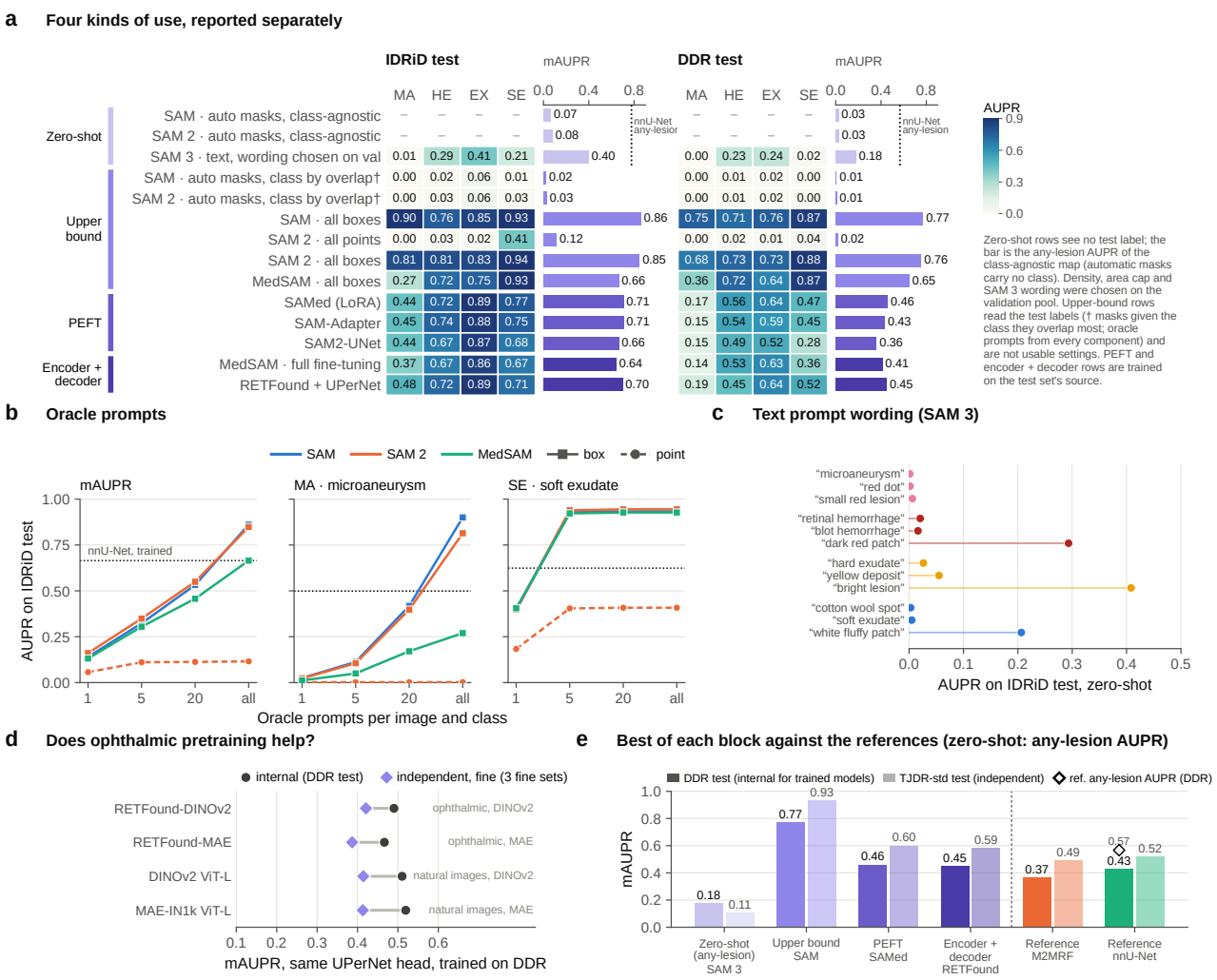


Fig. 5 Foundation models under four kinds of use. a, Per-class AUPR and mAUPR on the IDRiD and DDR test sets. Zero-shot rows see no test label and are summarised by the any-lesion AUPR of their class-agnostic maps (dotted line, nnU-Net on the same metric where measured); upper-bound rows read the test labels (automatic masks assigned the class they overlap most, prompts from every ground-truth component); the two trained settings are trained on the source of the test set. **b**, Oracle prompts: AUPR on IDRiD against the number of ground-truth boxes or points given per image and class; dotted line, nnU-Net trained on IDRiD. **c**, SAM 3 with three wordings per lesion class. **d**, Four ViT-L encoders under the same UPerNet head and recipe, trained on DDR (seed 0): internal test and mean of the three fine-grained independent test sets (the two TJDR sets and MAPLES-DR); lines join each retinal encoder to the natural-image encoder trained with the same objective. **e**, Best model of each block against the fundus-specific and self-configuring references; the zero-shot bar is an any-lesion AUPR, and diamonds give the references' any-lesion AUPR on the DDR test.

Table 6 Foundation models under four kinds of use. The kinds are reported in separate blocks and are not ranked against each other: zero-shot use sees no test label (its density, area cap and wording are chosen on the validation pool); the upper-bound block reads the test labels (masks classed by their overlap with the ground truth, prompts derived from every ground-truth component) and is not a usable setting; the two trained settings follow the unified protocol. AUPR on the IDRiD and DDR test sets. **Bold** and underlined mark the best and second-best rows of each column within a block.

			IDRiD test					DDR test				
Model	**Use**	**Backbone**	MA	HE	EX	SE	**mAUPR**	MA	HE	EX	SE	**mAUPR**
Zero-shot: no test label; density, area cap and wording chosen on the validation pool (summary = any-lesion AUPR‡)												
SAM	automatic masks, class-agnostic	ViT-H	–	–	–	–	.07	–	–	–	–	.03
SAM 2	automatic masks, class-agnostic	Hiera-L	–	–	–	–	.08	–	–	–	–	.03
SAM 3	text, wording chosen on validation	–	.01	.29	.41	.21	**.40**	.00	.23	.24	.02	**.18**
Upper bounds that read the test labels (not a usable setting)												
SAM	automatic masks, class by overlap†	ViT-H	.00	.02	.06	.01	.02	.00	.01	.02	.00	.01
SAM 2	automatic masks, class by overlap†	Hiera-L	.00	.03	.06	.03	.03	.00	.01	.02	.00	.01
SAM	oracle boxes, every component	ViT-B	**.90**	.76	**.85**	.93	**.86**	**.75**	.71	**.76**	.87	**.77**
SAM 2	oracle points, every component	Hiera-L	.00	.03	.02	.41	.12	.00	.02	.01	.04	.02
SAM 2	oracle boxes, every component	Hiera-L	.81	**.81**	.83	**.94**	.85	.68	**.73**	.73	**.88**	.76
MedSAM	oracle boxes, every component	ViT-B	.27	.72	.75	.93	.66	.36	.72	.64	.87	.65
Parameter-efficient fine-tuning (trained on the source of the test set)												
SAMed	LoRA	ViT-B	.44	.72	**.89**	**.77**	**.71**	**.17**	**.56**	**.64**	**.47**	**.46**
SAM-Adapter	adapters	ViT-H	**.45**	**.74**	.88	.75	.71	.15	.54	.59	.45	.43
SAM2-UNet	adapters + U-shaped decoder	Hiera-L	.44	.67	.87	.68	.66	.15	.49	.52	.28	.36
Encoder + decoder (trained on the source of the test set)												
MedSAM	full fine-tuning	ViT-B	.37	.67	.86	.67	.64	.14	**.53**	.63	.36	.41
RETFound	MAE encoder + UPerNet	ViT-L	**.48**	**.72**	**.89**	**.71**	**.70**	**.19**	.45	**.64**	**.52**	**.45**
M2MRF	reference, Table 4	–	.45	.63	.87	.68	.66	.15	.41	.53	.39	.37
nnU-Net	reference, Table 4	–	.50	.67	.87	.62	.66	.16	.50	.61	.45	.43

‡The zero-shot block is summarised by the any-lesion AUPR (union of the four classes against the pixelwise maximum of the maps); automatic masks carry no class, so their per-class cells are empty, and SAM 3's per-class cells are its own text prompts. The any-lesion AUPR of the references on the same test sets (IDRiD / DDR): M2MRF – / –; nnU-Net 0.78 / 0.57 (– where the run predates the metric).

†Each automatic mask is assigned the class it overlaps most in the ground truth, an upper bound for class-agnostic proposals. Scores of automatic masks use the predicted IoU.

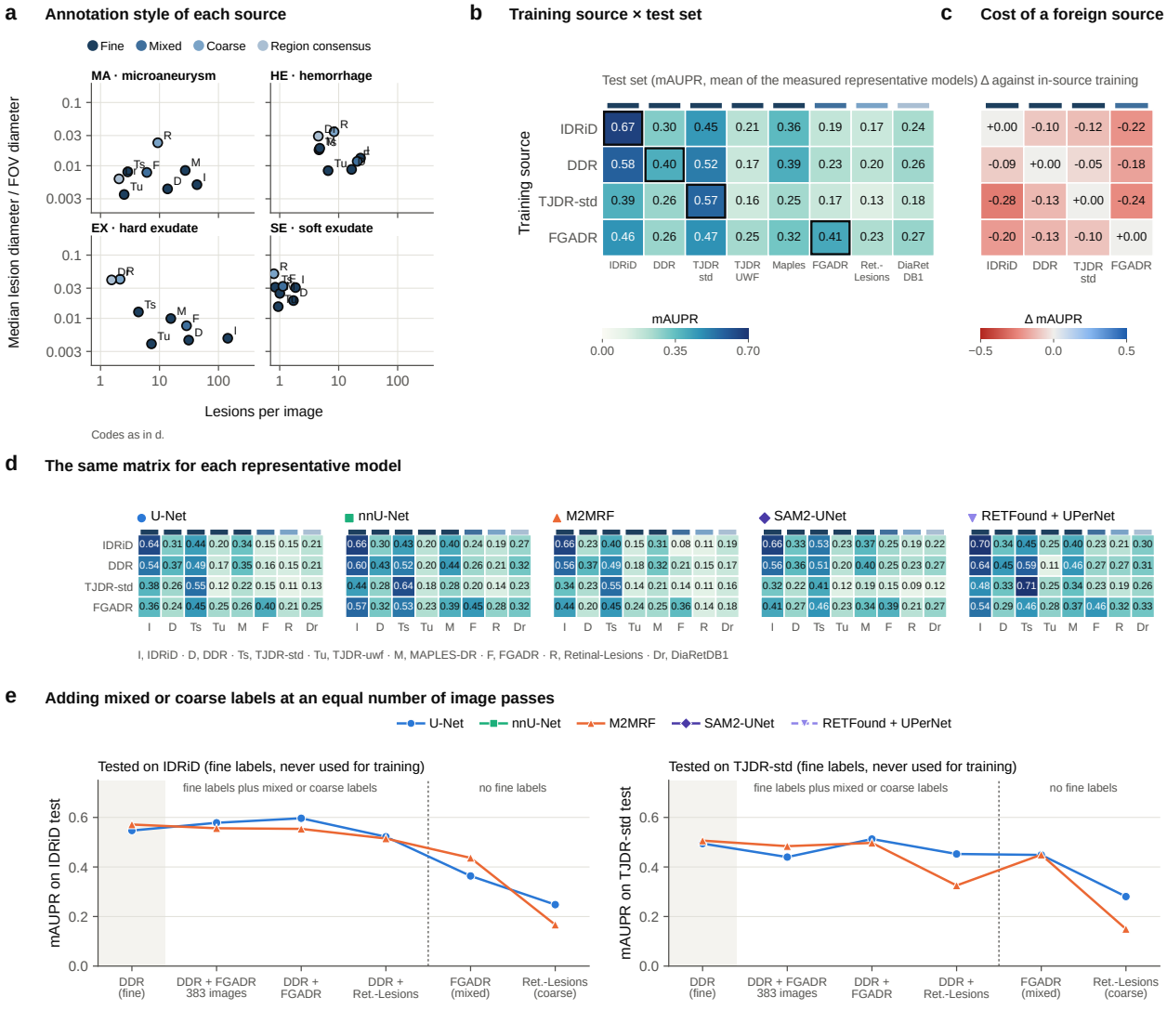
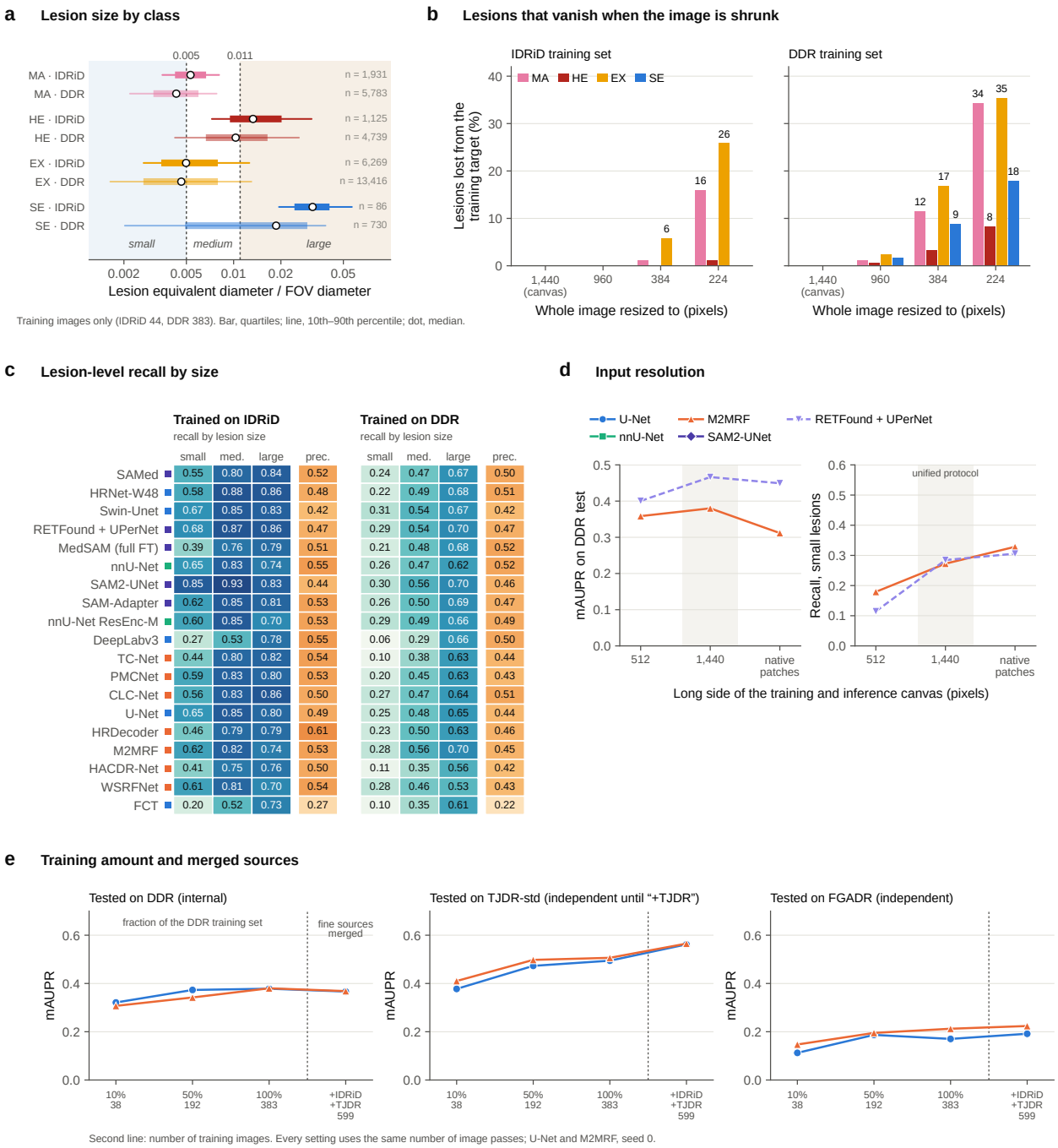


Fig. 6 Effects of annotation granularity and training source. **a**, Annotation style: connected components per image against their median diameter relative to the field-of-view width, per lesion class and source, over all images of each source (Methods). **b**, mAUPR for each training source (rows) and test set (columns), mean of five representative models, seed 0; outlined cells are internal tests. **c**, Difference to training on the source of the test set. **d**, The matrix of **b** for each model. **e**, Training sets of different granularity at an equal number of image passes, tested on two fine-grained test sets (U-Net and M2MRF, seed 0).



Training images only (IDRiD 44, DDR 383). Bar, quartiles; line, 10th–90th percentile; dot, median.

Second line: number of training images. Every setting uses the same number of image passes; U-Net and M2MRF, seed 0.

Fig. 7 Effects of lesion size, input resolution and training amount. a, Lesion diameter relative to the field-of-view diameter on the training images; dashed lines, the frozen boundaries between small, medium and large lesions. **b**, Share of connected components that vanish from the training target when the whole image is resized to the given size. **c**, Lesion-level recall by lesion size and lesion-level precision on the internal tests, at the threshold chosen on the validation set. **d**, mAUPR and recall of small lesions on DDR against the long side of the canvas (M2MRF and RETFound with UPerNet, seed 0). **e**, mAUPR against the fraction of the DDR training set and after the fine-grained sources are merged in (U-Net and M2MRF, seed 0); the number of image passes is the same in every setting.

Table 7 One factor at a time. One or two representative models are retrained with a single factor changed and everything else fixed to the unified protocol (seed 0). Each cell is the value followed by the change against the reference row of its block; – marks a model the block does not use. The last column averages the models of the block; its shading follows the change. Differences smaller than twice the median seed standard deviation of the main table are not interpreted.

Setting	U-Net	M2MRF	RETFound+UPer.	Mean
Annotation granularity of the training set — mAUPR on IDRiD test, equal image passes; U-Net and M2MRF				
DDR ■ (ref.)	.55	.57	–	.56
DDR + FGADR, 383 images ■■	.58 ▲.03	.56 ▼.02	–	.57 +.01
DDR + FGADR ■■	.60 ▲.05	.55 ▼.02	–	.58 +.02
DDR + Retinal-Lesions ■■	.52 ▼.03	.51 ▼.06	–	.52 −.04
FGADR ■	.36 ▼.18	.44 ▼.13	–	.40 −.16
Retinal-Lesions ■	.25 ▼.30	.17 ▼.40	–	.21 −.35
Input resolution, long side in pixels — mAUPR on DDR test; M2MRF and RETFound + UPerNet				
512	–	.36 ▼.02	.40 ▼.07	.38 −.04
1,440 (ref.)	–	.38	.47	.42
native-resolution patches	–	.31 ▼.07	.45 ▼.02	.38 −.04
Input resolution — lesion-level recall of small lesions on DDR test				
512	–	.18 ▼.09	.11 ▼.17	.15 −.13
1,440 (ref.)	–	.27	.28	.28
native-resolution patches	–	.33 ▲.06	.31 ▲.02	.32 +.04
Training amount — mAUPR on DDR test; U-Net and M2MRF				
DDR, 10% (38 images)	.32 ▼.06	.31 ▼.07	–	.31 −.07
DDR, 50% (192)	.37 ▼.01	.34 ▼.04	–	.36 −.02
DDR, 100% (383) (ref.)	.38	.38	–	.38
+ IDRiD + TJDR standard field (599)	.37 ▼.01	.37 ▼.01	–	.37 −.01
Loss, architecture fixed — mAUPR on DDR test; U-Net				
cross-entropy (ref.)	.36	–	–	.36
batch Dice	.24 ▼.12	–	–	.24 −.12
cross-entropy + batch Dice (unified protocol)	.38 ▲.02	–	–	.38 +.02
cross-entropy + per-image Dice (registered recipe)	.12 ▼.24	–	–	.12 −.24

Granularity of the labels used for training: ■ fine ■ mixed ■ coarse. In the training-amount block the number of image passes is the same in every row. The FGADR row of the granularity block is the FGADR-trained model of the cross-source experiment.

Discussion

We assessed methods for segmenting diabetic retinopathy lesions in fundus photographs by auditing published numbers, scoring and rerunning the original models, retraining 19 models under one protocol, and testing all of them on sources they had never seen. The design separates three things that a single leaderboard number mixes: what a paper reports, what its code produces, and what its architecture contributes once data, budget and evaluation are equal.

Reproducibility (Q1). Most fundus-specific methods cannot be rerun: 4 of 21 have training code. Where code and weights exist, the authors' evaluation code reproduced the released numbers exactly, so the distance between a paper and our protocol is a property of the metric and not of the model; but the released training recipe, run with the authors' settings on three seeds, produced models 5–7 points below the released weights. The literature audit shows why published tables cannot be read as rankings: most papers do not say how the AUPR was computed, the fixed-threshold implementation adds a model-dependent amount, and nearly half of the baseline numbers (22% identical to the method's own paper, 26% to an earlier copy) were carried from one paper to the next, at times with transposed columns.

Comparison (Q2). Under equal conditions two adapted foundation models (a low-rank adaptation of SAM and RETFound with a segmentation head) and the general-purpose HRNet-W48 were at the top of both internal tests, SAM-Adapter, Swin-Unet and the self-configuring nnU-Net followed within 0.04, and the seven fundus-specific networks were below them, in line with what has been found for three-dimensional medical segmentation[24]; over the eight test sets 11 of the 19 ranked models could not be separated from the first. That statement holds for the budget and the 1,440-pixel canvas of the unified protocol: the strongest fundus-specific method, HRDecoder, matched the leading models when run with its authors' released weights, which were trained at twice the resolution and on several times the pixels, and a lesion-specific decoder may need exactly the conditions a shared protocol takes away. The differences among the leading models were of the size of the seed-to-seed variation, whereas the loss of a single rare class, or a recipe that does not converge within the budget, moved a model by ten times as much. Outside the training source the order was less stable than the internal tables suggest: internal rank predicted independent rank on the sources whose lesions are drawn the same way, and the lead on coarsely labelled, region-labelled or ultra-widefield images went to models from the lower half of the internal tables. A model chosen on IDRiD or DDR alone is therefore chosen for the labelling style of that source as much as for the lesions. Where one score is needed, averaging the probability maps of several retrained models gained 0.02–0.06 mAUPR over the best single model on every fine-grained test set and -0.01 to 0.02 on the coarse and region-labelled ones, more than the differences among the leading architectures; the four adapted foundation models averaged by themselves gave the same gain. On the one source with several independent expert annotations, the agreement of the better models with the expert consensus was within the agreement among the experts, with the caveats that a region consensus is a coarse ground truth and that the experts' confidence maps give only a four-point curve. The strongest single models of the study were not in the ranked table. Under one decoder head, one recipe and one training amount, RETFound's two encoders scored no higher than the natural-image encoders they were initialised from (-0.053 and -0.020 mAUPR on DDR, -0.021 and -0.028 on IDRiD, one seed each), and the ImageNet masked autoencoder and DINOv2, ViT-L encoders that have never seen a fundus photograph, reached 0.52 and 0.51 on DDR and 0.73 and 0.74 on IDRiD under that head, above every ranked model on both sources. These cells were trained as controls after the ranked rows had been registered, each with one seed and one learning rate, so they do not enter the ranks; but they say that the lead of a ViT-L encoder with a UPerNet head over the convolutional and fundus-specific networks is a lead of the architecture and of the scale of pretraining behind the encoder, which retinal pretraining did not add to, and that the leaderboard of this task would be headed by a generic encoder and a generic decoder if such a configuration were registered. Confirming this with more seeds and a learning-rate sweep is the first item of future work. Zero-shot use of promptable models is not a baseline for this task, and results obtained with prompts from the ground truth should be labelled as upper bounds.

What decides performance (Q3). The largest effects came from the data and the protocol rather than from the architecture. Whether the test images share the annotation style and the field of view of the training images moved mAUPR by 0.16 on average and up to 0.28 between the four training sources, as much as the differences between the architectures of the main table: models trained on DDR kept or improved their score on the other pixel-labelled standard-field sources and lost a third to a half of it where the labels are coarse or regional or the field is ultra-wide, and models trained on the 44 IDRiD images lost on every other source, as Playout and Cheriet observed for U-Net-type baselines[22]. Mixed or coarse labels added to fine ones did not help and often hurt, and the most useful foreign source was the largest fine-grained one rather than the largest one. Shrinking the canvas from 1,440 to 512 pixels cost up to 0.07 mAUPR and between a third and three fifths of the small lesions, more images of the same source helped with diminishing returns, and merging the fine-grained sources improved the scores outside them. An internal test therefore says little about new images unless they are labelled like the training set, a conclusion shared with assessments in other fields[26].

Recommendations. For developers of methods: (1) use the official split, and publish the list of image identifiers when a new split is needed; (2) report the exact pooled AUPR at the original resolution inside the field of view, and name the implementation; (3) rerun every baseline or state for each number where it was taken from; (4) include at least one test set from a source that was not used for training, and do not average across annotation granularities; (5) include a self-configuring baseline and report several seeds; (6) release probability maps, so that any metric can be recomputed. For users: the lesion-level recall for small lesions and the result on an independent source with the user's field of view are more informative than a mean AUPR on IDRiD.

Limitations and scope. The scope of the study was fixed before the factor experiments were run, to what one protocol and one cluster could carry: the main table has three seeds for nine of its 19 configurations and one seed for the others, the factor experiments (training source, granularity, resolution, training amount, loss) use one to five representative models and one seed and report only differences larger than the seed variation of the main table, the encoder comparison has one seed and one learning rate per cell, and the source-recipe track was stopped before any run finished. Bootstrap intervals over test images are given per class (Supplementary Table 4); an interval for the class mean, from the same replicates, is left for the final version, and on both internal tests the per-class intervals of the leading models overlap. Left for future work are the remaining seeds and models of each factor experiment, a registered ranked row for the natural-image encoders under the UPerNet head and the full encoder grid (supervised ImageNet, frozen encoders, learning-rate sweeps, the adaptation-by-initialization square), the second-tier and rewritten models (UNet++, UPerNet, PerSAM, H2Former, L-Seg, SAA, RTNet), the released WSRFNet weights, vessel maps and text as additional inputs, the intermediate settings of each factor (a 1,024-pixel canvas, a quarter of DDR, merging all sources, the bin loss), and the test sets that were not available at the time (e-ophtha, ROC, ultra-widefield OCTA). Region-consensus and coarse ground truth bias pixel metrics, which is why DiaRetDB1 is reported with its image-level protocol and Retinal-Lesions as a separate column. The licence of FGADR does not allow derived data to be released, and we do not release them for Retinal-Lesions either. Most sources carry no patient identifiers, so leakage can be audited only through near-duplicate images. The ultra-widefield test set comes from a single centre and device. IDRiD has 27 test images, and intervals on it are wide. The first round used the hyperparameters of each source recipe, with a small learning-rate sweep on DDR as the only tuning. H2Former, although registered in tier 1, did not fit the cards used here at its native input size and is not in the main tables. Finally, the frozen lesion-matching rule is permissive (any overlap counts), so lesion-level precision is reported together with a stricter variant in the supplementary tables.

Methods

Workflow of model evaluation

The study has three parts that correspond to the three questions. (Q1) Published numbers are audited (the paper audit and the reproducibility funnel), released checkpoints are scored on official and independent test sets, and the authors' pipelines are rerun (author-recipe reruns). (Q2) Models with complete training code are retrained under one protocol on IDRiD and on DDR (unified retraining), including nnU-Net through its own entry point and foundation models under four kinds of use; all checkpoints are applied to six independent test sets (independent tests). (Q3) One factor is varied at a time on representative models (U-Net, nnU-Net, M2MRF, SAM2-UNet, RETFound with a UPerNet head): training source, annotation granularity, input resolution, training amount and loss; the HRDecoder case study of the main text applies the same machinery to one model's own geometry. Computational cost is recorded for every run. Each experiment is named in the text by what it does; the identifiers under which the runs, the results table and the planning documents of the repository file them are listed in Supplementary Table 18. The evaluation protocol and all splits were committed to the repository before the first main-table run. Pipeline checks had scored a reference U-Net and the released M2MRF checkpoints on the IDRiD and DDR test sets before that; the field-of-view rule and the lesion-size boundaries were then revised once, from training images only. After the freeze, the training amount and the selection rule were set from training loss and validation curves, and no test metric was used. One exception is disclosed in Supplementary Note 2: the source recipe of the reference U-Net was revised after the test score of a smoke run had been seen; this affects the source-recipe track only.

Data collection

Training sources. IDRiD[3] provides 81 images of 4288×2848 pixels with pixel masks for MA, HE, EX and SE; the official split is 54 training and 27 test images. A missing mask file means that the image has no lesion of that class. DDR[5] provides 757 images with the official split 383 / 149 / 225.

Independent test sources. TJDR[8] contains 561 images from one hospital: 257 standard-field images (2048×2048, 35–50°) and 304 ultra-widefield images (3912×3912, 133°), annotated with polygons by two ophthalmologists independently and adjudicated by a third; the four classes are mutually exclusive. The two fields are treated as two datasets. MAPLES-DR[42] adds pixel labels for 198 MESSIDOR[43] images (45°; 1440×960 to 2304×1536 pixels): seven retinologists drew

MA, HE, EX and SE at 1500×1500 on the field-of-view crop, MA, HE and EX starting from a network's pre-segmentation and SE from scratch; the labels are placed back on the original image, and only the official test split of 60 images is used. The FGADR Seg-set[6] contains 1,842 images of 1280×1280 pixels with masks for the four classes and, as extension labels, intraretinal microvascular abnormalities (159 images) and neovascularization (49). Retinal-Lesions[7] (release 20191227) contains 1,593 images from EyePACS, cropped to the field of view and resized to 896×896, from 1,505 patients, annotated with ellipses and polygons by at least three of 45 graders. DiaRetDB1 v2.1[21] contains 89 images of 1500×1152 pixels (50°) in which four experts independently marked regions (4,135 circles, 763 ellipses, 1,036 polygons), each with a confidence level.

Granularity tiers. Sources are assigned to four tiers: fine (IDRiD, DDR, TJDR standard field, TJDR ultra-widefield, MAPLES-DR), mixed (FGADR), coarse (Retinal-Lesions) and region consensus (DiaRetDB1). Results are averaged only within a tier.

Label mapping. Only the four classes are mapped; other classes are stored as extension labels and do not enter the four-class mean. TJDR: palette 1, 2, 3, 4 to EX, HE, MA, SE. FGADR: HardExudate, Hemohedge, Microaneurysms, SoftExudate. Retinal-Lesions: hard_exudate, retinal_hemorrhage, microaneurysm, cotton_wool_spots; vitreous and preretinal hemorrhage are not merged into HE. DiaRetDB1: Hard_exudates, Haemorrhages, Red_small_dots, Soft_exudates. MAPLES-DR: Microaneurysms, Hemorrhages, Exudates, CottonWoolSpots; its uncertain red lesions (MA or HE) and uncertain bright lesions (EX, SE or drusen) are ignored for the two classes they could belong to, and drusen and neovascularization are extension labels.

Model collection

The literature search was frozen at 30 September 2026. A fixed query (three concepts in title or abstract: diabetic retinopathy; lesion, microaneurysm, exudate, hemorrhage or cotton wool; segmentation) was run on PubMed, Europe PMC and OpenAlex for 2015 onwards. It identified 1,389 records, of which 948 passed the Boolean gate and 764 remained after de-duplication by title; 112 entered the full-text queue. Screening of a priority list (27 DOIs, one dataset lead and one further paper found when a mis-assigned DOI was corrected) required colour fundus photographs, pixel-level segmentation of at least two of the four classes and an obtainable public dataset; it included 24 methods and 2 datasets and excluded 3 records. Together with a snowballed corpus the ledger holds 86 records. Only the authors' own repositories count as official code.

39 model variants were registered in three tiers (22, 6 and 11). Each has a reproducibility level: R0, no code; R1, model definition only; R2, inference code and weights; R3, training code; R4, training code, weights, splits and environment. Each is assigned to evaluation tracks: A, author weights; B, unified retraining of the official code; C, re-implementation from the paper, marked in the tables; a re-implementation fixes every unstated choice in a written list of assumptions before any experiment and is not tuned to the published numbers, and its parameter count is compared with the published one and reported. Methods whose repository holds the model but no training loop (WSRFNet) or whose code cannot run on current hardware (PMCNet, TensorFlow 1.13) are retrained under Track B with the loss and optimizer the paper states, written into the wrapper and disclosed as such. One method with official code and weights, HRDecoder, was missed by the frozen search and added on 14 October 2026 after a targeted check for code-available methods; the addition is recorded as a protocol deviation. Authors of methods without code received a fixed e-mail on 3 October 2026 asking for code, weights and splits, with a deadline of four weeks; where code exists without weights, the weights are requested separately. Non-responses and failures remain in the denominator of the reproducibility funnel.

Preprocessing

Unified layout. Every source is exported to the same layout: the image at original resolution with the bytes of the release, one binary mask per class, a field-of-view mask and the split lists. A missing mask file is an all-negative image for that class; the image stays in the split.

Field of view. The field-of-view (FOV) mask is the largest connected component of pixels with luminance $0.299R + 0.587G + 0.114B > 10$, with holes filled. An Otsu threshold was abandoned after it removed the darker half of the retina in several training images. With the fixed threshold, the largest share of lesion pixels lying more than 2 pixels outside the mask on the training images is 2.3×10^{-5} (IDRiD hemorrhages) and 0 on DDR. For the independent sources one rule is added: if the four 40×40 corner blocks are uniform (1st to 99th percentile within 5 grey levels), the threshold is the larger of 10 and the corner median plus 3. It changes one of 4,085 images, a DiaRetDB1 image with a uniform grey background.

Canvas. For training and inference the image is cropped to the FOV and resized so that its long side is 1,440 pixels. Predictions are mapped back to the original resolution by bilinear interpolation, set to zero outside the FOV and stored as 16-bit images (probability = value / 65535). All metrics are computed offline from these maps.

Source-specific handling. FGADR masks have anti-aliased edges with values 1–254; a pixel is positive at ≥ 128, and the number of pixels dropped is recorded per class. In Retinal-Lesions, value 255 is lesion and value 127, which the authors use for annotations that do not follow the grading guideline, is written to a per-class ignore region that is skipped by all metrics and, when this source is used for training, by the loss. In DiaRetDB1, circles, ellipses and polygons are rasterized at pixel centres; for one expert and class the highest confidence per pixel is kept, with High = 1.0, Medium = 0.75 and Low = 0.25; the four experts are averaged and a pixel is positive at ≥ 0.75, the threshold of the original protocol. With this rule six of the eight positive-image counts of the original paper are matched exactly and two differ by one image (the preparation report in the repository); the union of all expert marks is kept as a lenient ground truth for sensitivity analysis.

Exclusive training targets. Models with a softmax head need one label per pixel. Overlapping pixels are written in the order EX, HE, SE, MA, later classes overwriting earlier ones. This removes less than 0.5% of the positive pixels of any class in the training sets (at most 0.34% at original resolution). Evaluation always uses the four original, possibly overlapping masks.

Construction of training, validation and test sets

Official splits are never changed. New splits use the generator PCG64 with seed 20261003, and the lists were committed before any training on the source concerned. Where a split is stratified, quotas per stratum follow the largest-remainder method. IDRiD: 10 of the 54 official training images, drawn at random from the sorted identifiers, are held out for validation (44 / 10 / 27); author-recipe retraining uses all 54. DDR: official 383 / 149 / 225. TJDR: the official test set is kept and 15% of the official training images of each field are drawn for validation, giving 172 / 30 / 55 (standard) and 209 / 37 / 58 (ultra-widefield). FGADR: stratified by file-name suffix and DR grade, 30% test and 15% of the remainder for validation (1,096 / 193 / 553). Retinal-Lesions: split by patient, stratified by the higher grade of the two eyes (951 / 164 / 478). DiaRetDB1: official 28 / 61; the 28 training images are used only to choose thresholds.

Test scenarios and leakage audit

The internal test is the official test set of the training source. Independent tests come from sources excluded as a whole from training and from internal testing; conclusions rest on them, as in the template study, where independent databases were held out and similar sequences were removed[26]. Every main-table checkpoint is applied to the test splits of the five independent sources. Ultra-widefield images are predicted at their native resolution with sliding windows, because matching pixels per degree to the 1,440-pixel canvas would require upsampling; a version resized to 1,440 pixels is reported for comparison. The per-degree scale is defined by the field of view of the training source; for a run whose training sources do not fix one (the factor cells trained on the standard-field TJDR images, on FGADR, or on merged sources), only the 1,440-pixel version is run and stands for the ultra-widefield result. For author weights, a source that the original model was trained on is removed from its independent tests. Near-duplicate images are searched with perceptual hashes and feature similarity between sources, between the splits of a source, and between the pretraining sets of foundation models and the test sets; an independent test image that duplicates a training image is removed and listed (Extended Data Fig. 3).

Evaluation of the original models

Released checkpoints (author weights). Released checkpoints are run in a current framework; for M2MRF, whose repository requires a framework version that does not support the cards used here, the forward pass was checked by reproducing the repository numbers with the authors' function. Each checkpoint is reported with three numbers: the paper, the authors' evaluation function, and this study's protocol. The authors' function for M2MRF evaluates the whole image, overwrites overlapping labels in the order EX, HE, SE, MA and computes AUPR from 11 thresholds.

Author-recipe reruns. M2MRF and HRDecoder are retrained with the authors' pipelines, augmentation and training set (all 54 IDRiD training images; three M2MRF seeds and one HRDecoder run, plus a first M2MRF run with the wrong effective batch that is reported separately) and scored at the last iteration without checkpoint selection, with the authors' function and with this study's protocol. The registered plan had a DDR run and HACDR-Net's recipe as well; these were cut when the scope was fixed, and the second and third M2MRF seeds were run when cards were free. For M2MRF: binary Dice loss (smoothing 10^{-5}); SGD with learning rate 0.01, momentum 0.9 and weight decay 5×10^{-4}; polynomial decay with power 0.9; 40,000 iterations on IDRiD and 60,000 on DDR; random crops of 960×1440 after resizing; photometric distortion; HRNet-W48 initialized from ImageNet. For HRDecoder: the authors' segmentor and loss, images resized to 2880×1920 with scale jitter 0.5–2, random 1920×1920 crops, flips and rotations of $\pm 45^{\circ}$, SGD 0.01 with 1,500 warm-up steps and linear decay over 20,000 steps, four images per step; their four-process synchronized batch normalization becomes one process with four images. The authors' runner evaluates on the test split every 2,000 iterations and keeps the best checkpoint; the rerun reports the last step, and what that test-split selection would have returned is given separately.

Unified retraining

What is shared. Data, splits, the 1,440-pixel canvas, augmentation, the training amount in pixels, checkpoint selection, inference geometry and evaluation are the same for every model. Each model keeps its loss, optimizer, learning rate, weight decay, schedule shape and effective batch from its source recipe (Table 8); the schedule is stretched over the shared number of steps.

Training amount. One equivalent epoch draws as many patch pixels as the canvases of the training split contain, so that a model trained on 224×224 patches draws about 18 times more patches per epoch than one trained on 960×960 patches:

$$\text{steps per epoch} = \left\lceil \frac{1}{B} \left\lceil \frac{\sum_i H_i W_i}{h\,w} \right\rceil \right\rceil, \tag{1}$$

where $H_i \times W_i$ is the canvas of training image i, $h \times w$ the patch and B the effective batch. The number of equivalent epochs was fixed per dataset by a budget probe: U-Net and M2MRF were trained once (seed 0) at twice a provisional amount, and only training loss and validation mAUPR were read. The validation curve was smoothed with a centred three-point mean; if the smoothed maximum of any probe of a dataset lay in the last 10% of the steps, the amount was doubled. This gave 1,000 equivalent epochs for IDRiD and 400 for DDR. The rule was revised twice after the validation curves had been seen and never after a test metric had been read; all versions of the rule and their outcomes are given in Supplementary Note 2.

Patches and inference. Models with a fixed input size (Swin-Unet and RETFound with UPerNet, 224; FCT, 384; CLC-Net, TC-Net and SAMed, 512; SAM-Adapter and MedSAM, 1,024) are trained on patches of that size cut from the canvas and predict with sliding windows of the same size at an overlap of 0.5; HRDecoder keeps its authors' windowed inference at 960. All other models are trained on 960×960 patches and predict on the whole canvas (Table 8).

Augmentation. Random scaling by 0.5–2 before cropping, horizontal flip with probability 0.5, rotation by multiples of 90°, and photometric distortion with probability 0.5 (brightness ± 32, contrast and saturation 0.5–1.5, hue $\pm 18°$). The randomness of a sample depends only on the seed and the sample index.

Checkpoint selection. A run is validated 20 times at equal intervals and the checkpoint with the highest validation mAUPR inside the FOV is scored; the last step is kept for a sensitivity analysis. A model whose best checkpoint falls in the last two validations is flagged as possibly under-trained. The test split is not read during training.

Reproducibility. A resume file stores model, optimizer, step, selection state, random-number state and data order; a run resumed after an interruption is bit-identical to an uninterrupted one. Each run writes its environment (host, card, driver, framework, seed, commit at process start, parameter count, wall time) and its recipe with every deviation.

Rare-class collapse. The reference U-Net uses a binary cross-entropy plus a sigmoid Dice term computed per image and class with smoothing 1. On a patch without positives this term is minimized by an all-zero output with a gradient of about 1 per pixel, whereas on a patch with a few positive pixels the gradient per pixel is about $2/(\text{number of positive pixels} + 1)$. Under patch training 83% of DDR patches and 67% of IDRiD patches contain no soft exudate. In the budget probe the soft exudate channel collapsed to zero on IDRiD, and on DDR the microaneurysm, hemorrhage and soft exudate channels did. With a smoothing of 10^{-5} the gradient of this term on an empty patch is small while the channel's output is far from zero, and a Dice pooled over the batch (nnU-Net) is not affected. The U-Net's Dice term was therefore pooled over the batch per class with smoothing 10^{-5}, as in nnU-Net; the revision was decided from the training and validation curves of a diagnostic run (validation mAUPR 0.569 on IDRiD and 0.544 on DDR against 0.127 before) after the registered recipe's test numbers had been read, and both versions are reported, the registered one in Supplementary Table 6 (Supplementary Note 2). Before any foundation model was trained the same rule was registered for SAM2-UNet, whose weighted IoU term is computed per image and class with smoothing 1, and for MedSAM, whose Dice term is computed per image and class with squared predictions and smoothing 10^{-5}: if the validation AUPR of a rare class stays at its prior level throughout training, the term is pooled over the batch and the model is retrained from its initial weights. The rule applied to SAM2-UNet on DDR (microaneurysm and soft-exudate validation AUPR 0.02–0.04 and 0.05–0.10 from the first to the last evaluation) and to MedSAM on DDR (microaneurysm 0.000–0.004 and soft exudate 0.002 at every evaluation, their prior levels, while hemorrhage and exudate trained); both models were retrained with the term pooled over the batch (SAM2-UNet to 0.664 and 0.360 mAUPR on IDRiD and DDR, MedSAM to 0.642 and 0.413), the main tables show the retrained runs, and the collapsed runs are kept as the registered recipe in the supplementary tables.

nnU-Net. nnU-Net is run through its own entry point in 2D at original resolution, with the same splits written into its split file and exclusive labels composed in the same overwrite order; softmax probabilities are exported, mapped to the original image and scored with the same evaluation. The default configuration and the residual-encoder preset M are both run. Its preprocessing is its own and is marked as such.

Source-recipe track. A second track, each model trained with its source recipe after minimal adaptation to the shared data, canvas and evaluation (seed 0), was registered to measure what moving a recipe to this task gives; it is not an architecture ranking, since the recipes differ about 67-fold in the number of images seen on IDRiD. Six of its runs were started and the track was stopped for compute on 2026-10-11 before any of them finished; no number from it is reported.

Foundation models

Four kinds of use are reported separately. (a) Zero-shot: automatic masks of SAM and SAM 2, scored once as class-agnostic "any lesion" proposals and once with each mask assigned the class it overlaps most; text prompts for SAM 3 with three wordings per class. (b) Oracle prompts: boxes or points generated from ground-truth connected components, with 1, 5, 10, 20 or all components per image and class; MedSAM, which accepts boxes only, belongs here. (c) Parameter-efficient fine-tuning: SAMed, SAM-Adapter and SAM2-UNet, trained on patches cut from the canvas (512, 1,024 and 960 pixels; Table 8). (d) Encoder plus decoder: MedSAM fully fine-tuned, and RETFound with a UPerNet head. To isolate the effect of ophthalmic pretraining, two pairs of ViT-L encoders fixed before any was trained are run on DDR with the same UPerNet head, recipe and training amount: RETFound's masked-autoencoder encoder against the ImageNet masked autoencoder it was initialised from, and RETFound's DINOv2 encoder against DINOv2 itself (one seed each; the pair differences are the pre-specified comparisons). After the DDR result of the first pair had been read, the three control encoders were also trained on IDRiD under the same head and recipe, so that both pairs exist on both training sources; this addition is reported as such. Settings (c) and (d) follow the unified protocol.

Evaluation of factors

Each experiment changes one factor and keeps the unified protocol otherwise; every cell is trained at the pixel budget of DDR with the DDR recipe of its model and one seed (seed 0), and differences smaller than twice the median seed standard deviation of the main table are reported without interpretation. *Training source*: the five representative models (U-Net, nnU-Net, M2MRF, SAM2-UNet and RETFound with UPerNet) are trained on the standard-field TJDR images and on FGADR and tested on all eight test sets, which with the main table gives four training sources. *Annotation granularity*: U-Net and M2MRF on six training sets at an equal number of patch pixels, namely DDR, FGADR, Retinal-Lesions, DDR with FGADR, DDR with Retinal-Lesions, and DDR with 383 FGADR images drawn once (list committed before training); the DDR and FGADR rows are the main-table and training-source runs. *Input resolution*: M2MRF and RETFound with UPerNet on DDR with a long side of 512 or 1,440 pixels (the main table) and with patches at native resolution; the HRDecoder case study of the main text trains that model at its authors' 2,880-pixel canvas and 1,920-pixel window with the equivalent epochs of the unified protocol counted at that canvas. *Training amount*: U-Net and M2MRF on 10%, 50% and 100% of the DDR training set (nested subsets drawn once) and on IDRiD, DDR and standard-field TJDR merged. *Loss*: U-Net on DDR with cross-entropy, batch Dice, their sum (the main table) and the per-image Dice of the registered recipe, with the architecture fixed. *Learning rate*: HRNet-W48 and M2MRF on DDR with the learning rate multiplied by 1/3 and 3, judged on validation mAUPR only. An extra vessel-map input and further resolution and training-amount levels of the registered plan were cut for compute and are listed as future work.

Metrics

Pixel level. For each class, the positive and negative pixels of all test images inside the FOV (and outside the ignore region of that class) are pooled, and AUPR is computed exactly over the 16-bit probability levels of the stored maps; this equals the average precision

$$\mathrm{AUPR} = \sum_k (R_k - R_{k-1})\, P_k, \tag{2}$$

with precision P_k and recall R_k at the k-th distinct threshold. mAUPR is the mean over the four classes. Dice and intersection over union are reported at a fixed threshold of 0.5 and at the per-class threshold that maximizes Dice on the validation set. For an image without positive pixels of a class the per-image Dice is missing, not zero.

Lesion level. Lesions are 8-connected components. A ground-truth lesion is detected if any predicted positive pixel overlaps it; a predicted component is a true positive if it overlaps any ground-truth pixel of the class. Precision, recall and F1 are pooled over the split. Lesion size is the equivalent diameter of the ground-truth component divided by the equivalent diameter of the FOV; the boundaries between small, medium and large, 0.005 and 0.011, are the medians of microaneurysms and hemorrhages on the 44 IDRiD and 383 DDR training images, rounded up to 0.001. Validation and test images were not used to set them.

Annotation style. The labelling style of each source (Fig. 6a) is measured on the prepared masks of all its images, every image once whatever its split: per class, the 8-connected components are counted and their equivalent diameter is divided by the long side of the FOV bounding box (which for a circular field equals the FOV diameter); ignore regions are removed before counting, and DiaRetDB1 is measured on the ≥ 0.75 consensus the protocol evaluates. Reported are the mean number of components per image, over all images of the source, and the median relative diameter over all components.

DiaRetDB1. The primary result follows the official image-level protocol: the image score of a class is the maximum probability inside the FOV, and the area under the ROC curve and the weighted error rates are reported. Pixel-level AUPR against the consensus regions is given separately and marked as region ground truth.

Authors' implementation. The 11-threshold AUPR is reported next to the exact value for all models, only to explain distances to published numbers.

Case study of one model's geometry. HRDecoder trains its backbone at half the resolution of a crop that is itself two thirds of the authors' 2,880-pixel canvas; the unified protocol halves the canvas and keeps that ratio. Three settings separate the recipe from the budget and the canvas: the authors' recipe retrained (the author-recipe rerun), the unified protocol at twice its training amount (for HRDecoder and, as a control, HRNet-W48 on DDR, trained on cards taken out of the queue), and the unified protocol on the authors' canvas with the authors' 1,920-pixel window and equivalent epochs counted at that canvas (four times the pixels per step). All three were registered before any of them was trained (docs/paper/14, note 46 of the repository).

Ensembles. Probability maps of the main-table models trained on one source are averaged pixel by pixel in five compositions fixed before any ensemble was scored: all models; the three and the five best by validation mAUPR; the best model of each family; and each family on its own. The member list is the 18 main-table rows scored when the compositions were registered; MedSAM, scored afterwards, is not a member. Members are chosen on validation data only; the ensemble's own validation maps set its thresholds; the test sets and the scoring are those of the single models. Ensembles are reported as a reference below the single models and do not enter the rank tests.

Reference model. As a starting point for users of the benchmark, two models are trained once on the merged training splits of the three fine-grained sources (IDRiD, DDR and the standard-field TJDR set; 599 images) with the recipe, canvas and pixel budget of their DDR main-table run, validated on DDR, and their probability maps are averaged as in the ensembles. The two were chosen before any merged run by their DDR validation mAUPR among the main-table models (RETFound with UPerNet, 0.643; SAMed, 0.632); the pair, the budget rule and the validation source were registered on 2026-10-23. The reference model is scored on every test set like the ensembles (thresholds from the DDR validation split of the averaged maps), is not a ranked row, and is released with its weights and maps and with a command that applies it to new images (`python -m bench.reference predict`: field-of-view detection by the rule above, the 1,440-pixel canvas, each member's sliding-window inference, the averaged probability map at the original resolution, and masks at the validation thresholds).

Human reference. On DiaRetDB1 each of the four experts in turn is scored as if it were a model: the expert's own markings, with the confidence mapping above, are the probability map, and the consensus of the other three (mean confidence ≥ 0.75) is the ground truth; thresholds come from the official training list as for the models, and the four values of each metric are reported as a band next to the DiaRetDB1 column. A map with four confidence levels gives a four-point precision–recall curve, so this band is a lower bound on what an expert's pixel-level AUPR would be with a continuous score. Lesion-level markings by three ophthalmologists on DDR and MAPLES-DR test images are pre-registered and will be reported in a Supplementary Note when complete.

Results by DR grade. For the test sets that carry a DR grade per image (DDR, FGADR, Retinal-Lesions, MAPLES-DR), every score is also computed within grades 0–1 and within grades 2–4 from the stored per-image histograms, with the thresholds of the full test; ungradable images enter neither stratum. Image-level detection summarizes the maximum probability inside the FOV against the presence of the class in the ground truth.

Label revision. Refined IDRiD[44] re-annotates the IDRiD images as one 1024×1024 grey-level label per image (one class per pixel: background, vitreous hemorrhage, retina, fovea, vessels, optic disc, hard exudates, IRMA, hemorrhages, neovascularisation, cotton-wool spots and microaneurysms) on a crop of the retinal region. For each of the 27 test images the crop is located in the original image by a coordinate-descent search over the four edges of the box, started from our FOV box, under two resampling hypotheses (the box stretched to the square, or padded to a square and resized uniformly), and accepted only when the normalised cross-correlation between the resampled original and the release reaches 0.97 (all 27: padded, 0.9988–0.9996). The official masks and the FOV are resampled through the recovered geometry with nearest-neighbour interpolation; cotton-wool spots are read as soft exudates and the other refined classes are ignored. Each IDRiD-trained checkpoint's original-resolution test maps are moved through the same geometry and scored by the protocol evaluator against both label sets, with the thresholds of its own validation run; no model is retrained or re-run. The per-class differences between the two label sets (pixels, components, intersection over union) are reported with the scores.

Statistics

Confidence intervals are 95% percentile intervals from 2,000 bootstrap replicates over test images (seed 0); each replicate pools the histograms of the drawn images and recomputes AUPR, and a replicate without positive pixels is dropped; the interval of mAUPR averages the four class values of the same replicates and drops a replicate in which any class has no positive pixel. Point estimates use the histograms at full 16-bit resolution and bootstrap replicates use per-image histograms coarsened to 4,096 bins. Two models are compared by a paired bootstrap on AUPR with shared image indices, and by a two-sided Wilcoxon signed-rank test on per-image Dice with Holm correction. Seed mean and s.d. are reported separately from bootstrap intervals. Rankings are compared across test sets with the Friedman test and the Nemenyi critical difference[45],

$$\mathrm{CD} = q_\alpha \sqrt{\frac{k(k+1)}{6N}}, \tag{3}$$

for k models and $N = 8$ test sets (two internal and six independent). Scores are the seed means of Table 4; in an independent block the score of a model is the mean over its IDRiD-trained and DDR-trained checkpoints; the DiaRetDB1 block uses the official image-level AUC on DiaRetDB1 (mean over the four classes). A version restricted to the fine-grained test sets is given in the repository's analysis notebook; the distribution of ranks under the image bootstrap[25] is left for the final version. Agreement between published and retrained rankings is measured by Kendall's τ.

Time and resource consumption

Training ran on NVIDIA A100 cards with 40 GB in a node of eight, one task per card, with a queue that assigns tasks from longest to shortest expected duration. Evaluation of author weights, the first author-recipe rerun, data export and all metric computation ran on a workstation (RTX 4060 Ti, 16 GB). Where a card cannot hold the effective batch of a recipe, gradients are accumulated and the deviation is recorded. Parameter count, floating-point operations, peak training memory, GPU hours and inference time per image are reported for every model (Extended Data Fig. 10).

Data availability

IDRiD, DDR and TJDR are public. FGADR and Retinal-Lesions are available from their authors on request. The FGADR licence forbids redistribution of the images and of any derived data; Retinal-Lesions is distributed under application terms, and we handle it in the same way. DiaRetDB1 is public. The MAPLES-DR labels are public (CC BY 4.0); the MESSIDOR images they annotate are available from ADCIS on request. We release all split lists, the protocol files and, for IDRiD, DDR, TJDR, MAPLES-DR and DiaRetDB1, the probability maps of every model (the maps hold no image data; DiaRetDB1 is distributed for research use). For FGADR and Retinal-Lesions we release split lists and code only.

Code availability

The benchmark code, including data export, training under the unified protocol, inference, the evaluation module and the scripts that produce every figure and table of this paper, is available at `https://github.com/skylynf/retiseg`. Official model repositories are referenced as pinned submodules and are not modified.

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Table 8 What each model keeps under the unified protocol. Data, splits, the 1,440-pixel canvas, augmentation, the training amount in pixels, checkpoint selection and evaluation are shared (Methods). Loss, optimizer, learning rate, weight decay, schedule shape and effective batch are taken from each model's source recipe; the schedule is stretched over the shared number of steps. Models with a fixed input size train on patches of that size and predict with half-overlapping windows; all others train on 960×960 patches and predict on the whole canvas. nnU-Net runs through its own entry point and keeps its own plan.

Model	Initialization	Head	Loss	Optim.	LR	WD	Schedule	Batch	Patch	Inference	Steps
U-Net	none	4 sigmoid	BCE + Dice[a]	Adam	1e-4	0	constant	4	960	whole canvas	21,000
DeepLabv3	ImageNet, ResNet-101	4 sigmoid	BCE	SGD	7e-3	1e-4	poly 0.9	16	960	whole canvas	6,000
HRNet-W48	ImageNet	4 sigmoid	BCE	SGD	1e-2	5e-4	poly	12	960	whole canvas	7,000
Swin-Unet	ImageNet, Swin-T	5 softmax	0.4 CE + 0.6 Dice	SGD	1e-2	1e-4	poly 0.9	24	224	sliding window	63,000
FCT	none	4 sigmoid	BCE	Adam	1e-3	0	constant	4	384	sliding window	129,000
nnU-Net	none	5 softmax	CE + batch Dice	SGD	1e-2	3e-5	poly 0.9	2	1024×1536[c]	sliding window	250,000[c]
nnU-Net ResEnc-M	none	5 softmax	CE + batch Dice	SGD	1e-2	3e-5	poly 0.9	13[c]	384×640[c]	sliding window	250,000[c]
M2MRF	ImageNet, HRNet-W48	4 sigmoid	binary Dice	SGD	1e-2	5e-4	poly 0.9	4	960	whole canvas	21,000
HACDR-Net	none	5 softmax	weighted CE[b]	AdamW	6e-5	1e-2	poly 1.0, warm-up	1	960	whole canvas	83,000
HRDecoder	ImageNet, HRNet-W48	4 sigmoid	binary Dice + 0.1 Dice (HR crops)	SGD	1e-2	5e-4	poly 0.9	4	960	sliding window	21,000
PMCNet	ImageNet, EfficientNet-B0	5 softmax	CE	Adam	1e-4	0	poly 0.9	1	960	whole canvas	83,000
WSRFNet	ImageNet, EfficientNet-B1	4 sigmoid	binary Dice, per loop[d]	SGD	1e-2	5e-4	poly 0.9	4	960	whole canvas	21,000
CLC-Net[C]	none	5 softmax	Dice + BCE (cls. heads)[e]	Adam	3e-4	0	step (1/3, 5/6)	16	512	sliding window	73,000
TC-Net[C]	none	5 softmax	CE	Adam	5e-4	1e-5	constant	4	512	sliding window	73,000
SAMed	SAM ViT-B, LoRA r=4	5 softmax	0.2 CE + 0.8 Dice	AdamW	5e-3	0.1	warm-up, poly	24	512	sliding window	13,000
SAM-Adapter	SAM ViT-H, adapters	4 sigmoid	BCE + IoU	AdamW	2e-4	1e-2	stepped cosine	2	1,024	sliding window	37,000
SAM2-UNet	SAM 2 Hiera-L, adapters	4 sigmoid	BCE + weighted IoU[f]	AdamW	1e-3	5e-4	stepped cosine	12	960	whole canvas	7,000
MedSAM (full FT)	MedSAM ViT-B, all weights	4 sigmoid	Dice + BCE	AdamW	1e-4	1e-2	constant	2	1,024	sliding window	37,000
RETFound + UPerNet	RETFound ViT-L (MAE)	4 sigmoid	BCE + 0.4 aux.	AdamW	4.7e-4	0.05	warm-up, cosine	24	224	sliding window	63,000

Batch is the effective batch (gradient accumulation where one card cannot hold it). Steps is for IDRiD at 1,000 equivalent epochs; one equivalent epoch draws as many patch pixels as the training canvases contain, so steps = ⌈⌈canvas pixels / patch pixels⌉ / batch⌉ per epoch. DDR uses 400 equivalent epochs (for example 83,600 steps at batch 4 and 960-pixel patches; Supplementary Table 3 lists every run). [C], re-implemented from the paper. Every run's recipe record, with its deviations from the source recipe, is in the repository.
[a] Per-image, per-class sigmoid Dice with smoothing 1 in the registered recipe; the main-table runs pool the Dice term over the batch (Methods). The other Dice losses use 10^{-5}. [b] IDRiD: class-weighted cross-entropy; DDR: the same plus the authors' NALoss with weight 0.2. [c] nnU-Net plans patch and batch per dataset (IDRiD plans shown; DDR: default 1024×1536 at batch 2, ResEnc-M 512×512 at batch 12) and trains 1,000 epochs of 250 steps on both sources; its learning rate, weight decay and polynomial schedule are its defaults. [d] The paper's loss and optimizer, written into the wrapper because the repository has no training loop. [e] Local and contextual Dice plus binary cross-entropy of the two classification heads, as described in the paper. [f] Registered per-image term; the main-table run pools it over the batch (Methods).

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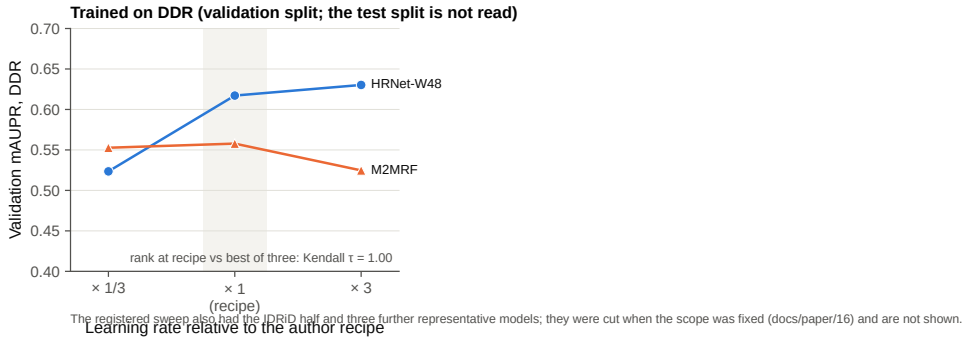
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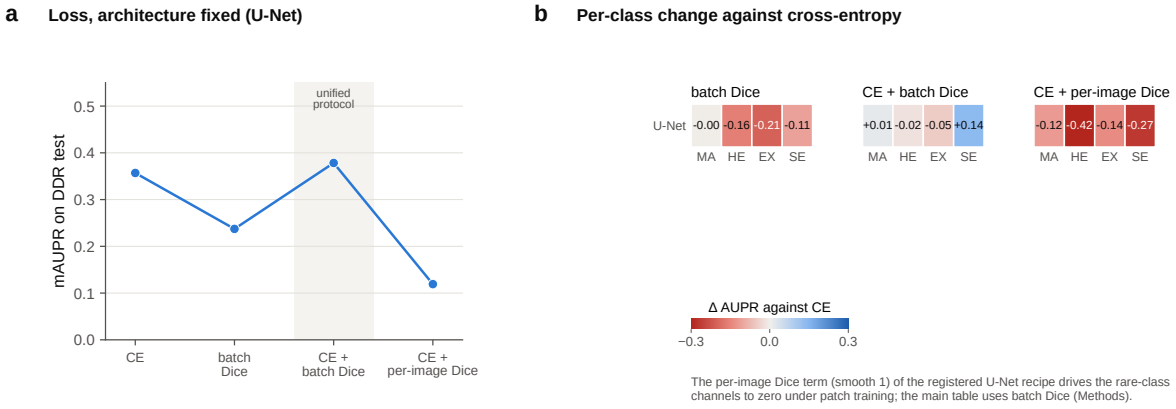
Extended Data

Ten Extended Data figures were planned; two were withdrawn (the source-recipe track stopped before any run finished; the per-image distributions duplicate what the released histograms give). Four are drawn below from the same data table as the main figures; the two that need images (1 and 2) are not yet drawn and will be made on the workstation that holds the data.

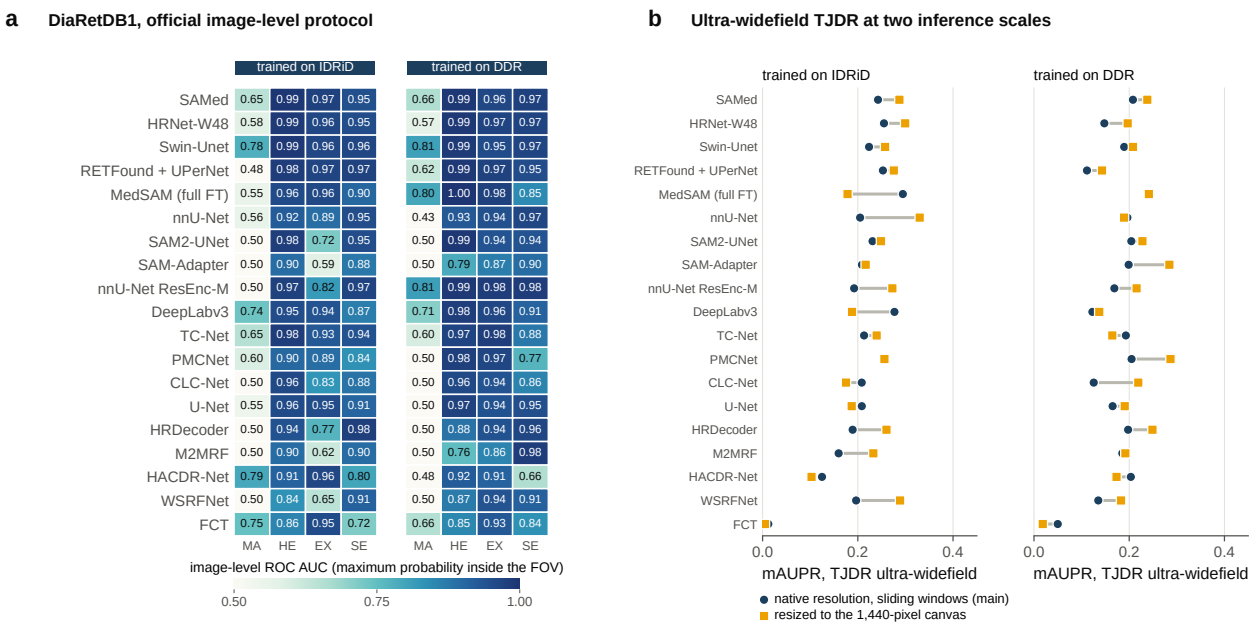
Extended Data Fig.	Content	Source	Status
1	Example images and labels of each source; annotation style	prepared datasets; label audit	needs images (workstation)
2	Field-of-view masks and canvas scale per source	FOV reports	needs images (workstation)
3	Leakage audit: near-duplicates between sources and splits	vessel-signature audit	done; reported as Supplementary Table 15
4	Sensitivity to the metric implementation (thresholds, FOV, overlap)	evaluation switches	with Fig. 2d,e
5	Per-image distributions and threshold sensitivity	per-image histograms	withdrawn (histograms stay in the repository)
6	(withdrawn; the source-recipe track was stopped before any run finished)	source-recipe track	–
7	Learning-rate sweep (DDR)	learning-rate sweep	drawn
8	Loss with the architecture fixed (U-Net)	loss experiment	drawn
9	DiaRetDB1 official protocol; ultra-widefield at two scales	independent tests	drawn
10	Computational cost	cost records of every run	drawn (cost audit pending)



Extended Data Fig. 7 Learning-rate sweep. Validation mAUPR on DDR of one general (HRNet) and one fundus-specific (M2MRF) network with the learning rate of the recipe multiplied by 1/3, 1 and 3. Kendall τ compares the order of the models at the recipe value with their order when each uses its best of the three values.

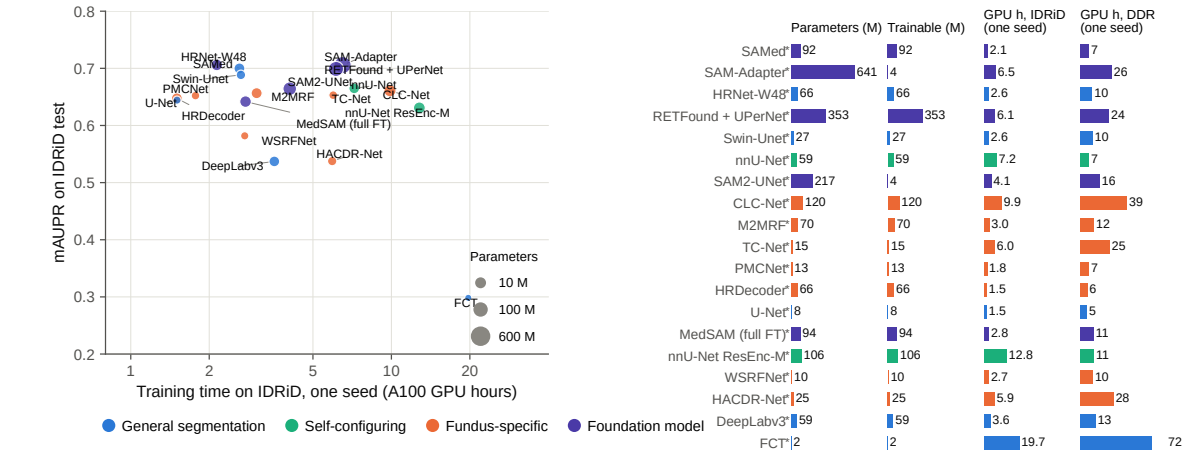


Extended Data Fig. 8 Loss with the architecture fixed. **a**, mAUPR of U-Net on the DDR test set with cross-entropy, batch Dice, cross-entropy with batch Dice (the main-table recipe) and cross-entropy with a per-image Dice term (the registered recipe, which collapses on rare classes); seed 0. **b**, Per-class change in AUPR against cross-entropy.



Models in the order of their average rank (Fig. 4e). Native resolution is the main-table setting; the resized version is the comparison variant registered for the independent tests.

Extended Data Fig. 9 DiaRetDB1 under its official protocol, and the ultra-widefield images at two inference scales. a, Image-level ROC AUC per lesion class on DiaRetDB1 (the maximum probability inside the field of view against the presence of the class in the ≥ 0.75 expert consensus), every main-table model from both sources, seed 0; a value of 0.5 means every image of the test set is positive or negative for the class. **b**, mAUPR on the ultra-widefield TJDR images predicted at native resolution with sliding windows (the main-table setting) and after resizing to the 1,440-pixel canvas.



Models ordered by mAUPR on the IDRiD test set. 40 GB A100, one task per card; nnU-Net follows its own 1,000-epoch schedule. * parameter count read from the run (the others are the plan's); FLOPs, peak memory and inference time join when the cost audit has run.

Extended Data Fig. 10 Computational cost. a, mAUPR on the IDRiD test set against the training time of one seed on IDRiD (40 GB A100, one task per card); marker area, number of parameters. **b**, Parameters, trainable parameters under the unified protocol and training time of one seed on each source; floating-point operations, peak memory and inference time join when the cost audit has run.

Supplementary information

Supplementary tables (separate document, generated from the same data table).

1. Literature audit: the setup stated by each method paper (split, resolution, AUPR implementation, baseline provenance).
2. One method, many reported values: every method reported in at least two papers on IDRiD or DDR, with the spread and the provenance of the values.
3. Training amount, batch and selection of every main-table run.
4. Per-class AUPR with 95% bootstrap intervals of every main-table model on every test set, both sources.
5. Secondary metrics on the internal tests: 11-threshold AUPR, Dice, IoU, lesion-level precision and recall by lesion size, image-level AUC.
6. Registered and revised recipes of the three models whose per-image loss term collapsed on rare classes.
7. Every factor cell, per class and per test set.
8. Computational cost.
9. Refined IDRiD: every IDRiD-trained model against the official and the refined labels, and the label differences.
10. Encoder comparison on both training sources.
11. Ensembles of every registered composition on every test set, and the reference model.
12. Results by DR grade.
13. Human reference on DiaRetDB1.
14. Annotation style of every source: components per image and their size.
15. Near-duplicate audit and the images excluded.
16. Released checkpoints on every test set, per class.
17. Foundation models without training on every test set.
18. Experiment names used in the text and the identifiers used in the repository.

Supplementary notes. Note 1, one entry per registered model variant (code used, reproducibility level, changes); Note 2, the training budget, the selection rule, the disclosed revisions and the scope changes; Note 3, the evaluation protocol.
Supplementary data. Split lists and exclusions; the literature extraction; the results table with every run's recipe and environment record; probability maps where the licence allows.